ALLEN CAREER INSTITUTE
KOTA (RAJASTHAN)
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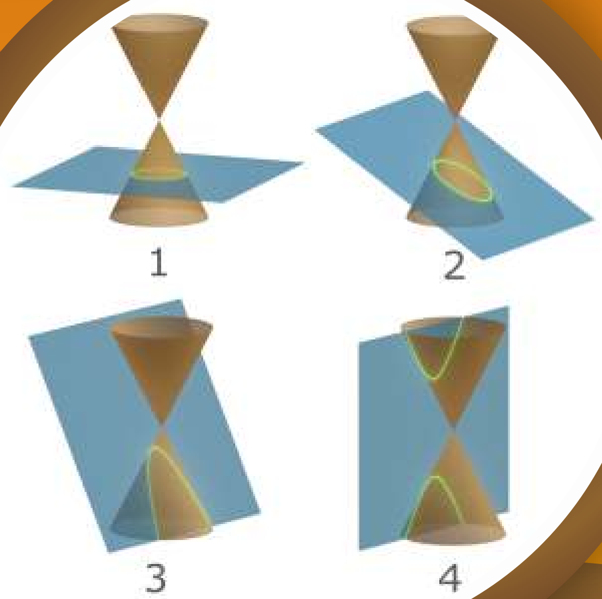
MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Conic Section

(English Medium)

EXERCISE (O-1)

Straight Objective Type

1. If the line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0$ then one of the values of 'k' is -

(A) 1/8 (B) 8 (C) 4 (D) 1/4 **PR0003**

2. The length of the intercept on y – axis cut off by the parabola, $y^2 - 5y = 3x - 6$ is

(A) 1 (B) 2 (C) 3 (D) 5 **PR0004**

3. A variable circle is drawn to touch the line $3x - 4y = 10$ and also the circle $x^2 + y^2 = 1$ externally then the locus of its centre is -

(A) straight line (B) circle

(C) pair of real, distinct straight lines (D) parabola **PR0006**

4. The locus of the point of trisection of all the double ordinates of the parabola $y^2 = \ell x$ is a parabola whose latus rectum is -

(A) $\frac{\ell}{9}$ (B) $\frac{2\ell}{9}$ (C) $\frac{4\ell}{9}$ (D) $\frac{\ell}{36}$ **PR0007**

5. The straight line $y = m(x - a)$ will meet the parabola $y^2 = 4ax$ in two distinct real points if

(A) $m \in R$ (B) $m \in [-1, 1]$

(C) $m \in (-\infty, 1] \cup [1, \infty)$ (D) $m \in R - \{0\}$ **PR0008**

6. If on a given base, a triangle be described such that the sum of the tangents of the base angles is a constant, then the locus of the vertex is :

(A) a circle (B) a parabola (C) an ellipse (D) a hyperbola **PR0010**

7. The equation of the circle drawn with the focus of the parabola $(x - 1)^2 - 8y = 0$ as its centre and touching the parabola at its vertex is :

(A) $x^2 + y^2 - 4y = 0$ (B) $x^2 + y^2 - 4y + 1 = 0$

(C) $x^2 + y^2 - 2x - 4y = 0$ (D) $x^2 + y^2 - 2x - 4y + 1 = 0$ **PR0011**

8. If a focal chord of $y^2 = 4x$ makes an angle $\alpha, \alpha \in \left(0, \frac{\pi}{4}\right]$ with the positive direction of x-axis, then minimum length of this focal chord is -

(A) $2\sqrt{2}$ (B) $4\sqrt{2}$ (C) 8 (D) 16 **PR0014**

9. A parabola $y = ax^2 + bx + c$ crosses the x-axis at $(\alpha,0)$ $(\beta,0)$ both to the right of the origin. A circle also passess through these two points. The length of a tangent from the origin to the circle is :

(A) $\sqrt{\frac{bc}{a}}$ (B) ac^2 (C) $\frac{b}{a}$ (D) $\sqrt{\frac{c}{a}}$ **PR0015**

10. If (2,–8) is one end of a focal chord of the parabola $y^2 = 32x$, then the other end of the focal chord, is-

(A) (32,32) (B) (32,–32) (C) (–2,8) (D) (2,8) **PR0016**

11. The length of a focal chord of the parabola $y^2 = 4ax$ at a distance b from the vertex is c, then

(A) $2a^2 = bc$ (B) $a^3 = b^2c$ (C) $ac = b^2$ (D) $b^2c = 4a^3$ **PR0018**

12. Locus of trisection point of any arbitrary double ordinate of the parabola $x^2 = 4by$, is -

(A) $9x^2 = by$ (B) $3x^2 = 2by$ (C) $9x^2 = 4by$ (D) $9x^2 = 2by$ **PR0019**

13. Consider the graphs of $y = Ax^2$ and $y^2 + 3 = x^2 + 4y$, where A is a positive constant and $x,y \in R$. Number of points in which the two graphs intersect, is-

(A) exactly 4

(B) exactly 2

(C) at least 2 but the number of points varies for different positive values of A.

(D) zero for atleast one positive A. **PR0020**

14. From an external point P, pair of tangent lines are drawn to the parabola, $y^2 = 4x$. If θ_1 & θ_2 are the inclinations of these tangents with the axis of x such that, $\theta_1 + \theta_2 = \frac{\pi}{4}$, then the locus of P is :

(A) $x - y + 1 = 0$ (B) $x + y - 1 = 0$

(C) $x - y - 1 = 0$ (D) $x + y + 1 = 0$ **PR0021**

15. y-intercept of the common tangent to the parabola $y^2 = 32x$ and $x^2 = 108y$ is

(A) – 18 (B) – 12 (C) – 9 (D) – 6 **PR0022**

16. The points of contact Q and R of tangent from the point P (2, 3) on the parabola $y^2 = 4x$ are

(A) (9, 6) and (1, 2) (B) (1, 2) and (4, 4)

(C) (4, 4) and (9, 6) (D) (9, 6) and $(\frac{1}{4}, 1)$ **PR0023**

17. If the lines $(y - b) = m_1(x + a)$ and $(y - b) = m_2(x + a)$ are the tangents to the parabola $y^2 = 4ax$, then

(A) $m_1 + m_2 = 0$ (B) $m_1m_2 = 1$

(C) $m_1m_2 = -1$ (D) $m_1 + m_2 = 1$ **PR0024**

18. The equation of the common tangent touching the circle $(x - 3)^2 + y^2 = 9$ and the parabola $y^2 = 4x$ above the x-axis is -

(A) $\sqrt{3}y = 3x + 1$ (B) $\sqrt{3}y = -(x + 3)$

(C) $\sqrt{3}y = x + 3$ (D) $\sqrt{3}y = -(3x + 1)$ **PR0026**

19. If x + y = k is normal to $y^2 = 12x$, then 'k' is-

(A) 3 (B) 9 (C) –9 (D) –3 **PR0028**

20. Equation of the other normal to the parabola $y^2 = 4x$ which passes through the intersection of those at (4,–4) and (9, –6) is -

(A) $5x - y + 115 = 0$ (B) $5x + y - 135 = 0$

(C) $5x - y - 115 = 0$ (D) $5x + y + 115 = 0$ **PR0029**

21. Length of the normal chord of the parabola, $y^2 = 4x$, which makes an angle of $\frac{\pi}{4}$ with the axis of x is:

(A) 8 (B) $8\sqrt{2}$ (C) 4 (D) $4\sqrt{2}$ **PR0030**

22. Suppose that three points on the parabola $y = x^2$ have the property that their normal lines intersect at a common point (a,b). The sum of their x-coordinates is -

(A) 0 (B) $\frac{2b-1}{2}$ (C) $\frac{a}{2}$ (D) a + b **PR0031**

23. The normal chord of a parabola $y^2 = 4ax$ at the point whose ordinate is equal to the abscissa, then angle subtended by normal chord at the focus is :

(A) $\frac{\pi}{4}$ (B) $\tan^{-1}\sqrt{2}$ (C) $\tan^{-1}2$ (D) $\frac{\pi}{2}$ **PR0032**

24. Which one of the following lines cannot be the normals to $x^2 = 4y$?

(A) $x - y + 3 = 0$ (B) $x + y - 3 = 0$

(C) $x - 2y + 12 = 0$ (D) $x + 2y + 12 = 0$ **PR0033**

25. Tangents are drawn from the points on the line $x - y + 3 = 0$ to parabola $y^2 = 8x$. Then the variable chords of contact pass through a fixed point whose coordinates are :

(A) (3,2) (B) (2,4) (C) (3,4) (D) (4,1) **PR0034**

26. Consider two curves $C_1 : \left(y-\sqrt{3}\right)^2 = 4\left(x-\sqrt{2}\right)$ and $C_2 : x^2 + y^2 = \left(6+2\sqrt{2}\right)x + 2\sqrt{3}y - 6\left(1+\sqrt{2}\right)$, then-

(A) C_1 and C_2 touch each other only at one point.

(B) C_1 and C_2 touch each other exactly at two points.

(C) C_1 and C_2 intersect (but do not touch) at exactly two points.

(D) C_1 and C_2 neither intersect nor touch each other. **PR0035**

27. C is the centre of the circle with centre (0,1) and radius unity. P is parabola $y = ax^2$. The set of values of 'a' for which they meet at a point other than the origin, is-

(A) $a > 0$ (B) $a \in \left(0, \frac{1}{2}\right)$ (C) $\left(\frac{1}{4}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \infty\right)$ **PR0037**

28. If the locus of the middle points of the chords of the parabola $y^2 = 2x$ which touches the circle $x^2 + y^2 - 2x - 4 = 0$ is given by $(y^2 + 1 - x)^2 = \lambda(1 + y^2)$, then the value of λ is equal to-

(A) 3 (B) 4 (C) 5 (D) 6 **PR0039**

29. PN is an ordinate of the parabola $y^2 = 4ax$(P on $y^2 = 4ax$ and N on x-axis). A straight line is drawn parallel to the axis to bisect NP and meets the curve in Q. NQ meets the tangent at the vertex in a point T such that AT = kNP, then the value of k is (where A is the vertex)

(A) 3/2 (B) 2/3 (C) 1 (D) none **PR0055**

30. Let A and B be two points on a parabola $y^2 = x$ with vertex V such that VA is perpendicular to VB and θ is the angle between the chord VA and the axis of the parabola. The value of $\frac{|VA|}{|VB|}$ is

(A) $\tan\theta$ (B) $\tan^3\theta$ (C) $\cot^2\theta$ (D) $\cot^3\theta$ **PR0056**

EXERCISE (O-2)

Multiple Correct Answer Type

1. The locus of the mid point of the focal radii of a variable point moving on the parabola, $y^2 = 8x$ is a parabola whose-

(A) Latus rectum is half the latus rectum of the original parabola

(B) Vertex is (1,0)

(C) Directrix is y-axis

(D) Focus has the co-ordinates (2,0) **PR0041**

2. Consider a circle with its centre lying on the focus of the parabola, $y^2 = 2px$ such that it touches the directrix of the parabola. Then a point of intersection of the circle & the parabola is

(A) $\left(\frac{p}{2}, p\right)$ (B) $\left(\frac{p}{2}, -p\right)$ (C) $\left(-\frac{p}{2}, p\right)$ (D) $\left(-\frac{p}{2}, -p\right)$ **PR0042**

3. The straight line $y + x = 1$ touches the parabola

(A) $x^2 + 4y = 0$ (B) $x^2 - x + y = 0$ (C) $4x^2 - 3x + y = 0$ (D) $x^2 - 2x + 2y = 0$

PR0045

4. P is a point on the parabola $y^2 = 4ax (a > 0)$ whose vertex is A. PA is produced to meet the directrix in D and M is the foot of the perpendicular from P on the directrix. If a circle is described on MD as a diameter then it intersects the x-axis at a point whose co-ordinates are :

(A) $(-3a,0)$ (B) $(-a,0)$ (C) $(-2a,0)$ (D) $(a,0)$ **PR0072**

5. If from the vertex of a parabola $y^2 = 4x$ a pair of chords be drawn at right angles to one another and with these chords as adjacent sides a rectangle be made, then the locus of the further end of the rectangle is -

(A) an equal parabola

(B) a parabola with focus at (9,0)

(C) a parabola with directrix as $x - 7 = 0$

(D) a parabola having tangent at its vertex $x = 8$

PR0073

6. PQ is a double ordinate of the parabola $y^2 = 4ax$. If the normal at P intersect the line passing through Q and parallel to axis of x at G, then locus of G is a parabola with -

(A) length of latus rectum equal to 4a

(B) vertex at (4a,0)

(C) directrix as the line $x - 3a = 0$

(D) focus at (5a,0) **PR0075**

7. TP and TQ are tangents to parabola $y^2 = 4x$ and normals at P and Q intersect at a point R on the curve. The locus of the centre of the circle circumscribing ΔTPQ is a parabola whose

(A) vertex is (1, 0).

(B) foot of directrix is $\left(\frac{7}{8}, 0\right)$.

(C) length of latus-rectum is $\frac{1}{4}$.

(D) focus is $\left(\frac{9}{8}, 0\right)$. **PR0076**

8. The locus of the point of intersection of those normals to the parabola $x^2 = 8y$ which are at right angles to each other, is a parabola. Which of the following hold(s) good in respect of the locus ?

(A) Length of the latus rectum is 2.
(B) Coordinates of focus are $\left(0, \frac{11}{2}\right)$
(C) Equation of a director circle is $2y - 11 = 0$
(D) Equation of axis of symmetry is $y = 0$

PR0077

9. Let A be the vertex and L the length of the latus rectum of the parabola, $y^2 - 2y - 4x - 7 = 0$. The equation of the parabola with A as vertex, 2L the length of the latus rectum and the axis at right angles to that of the given curve is :

(A) $x^2 + 4x + 8y \quad 4 = 0$
(B) $x^2 + 4x - 8y + 12 = 0$
(C) $x^2 + 4x + 8y + 12 = 0$
(D) $x^2 + 8x - 4y + 8 = 0$

PR0162

10. The locus of the mid point of the focal radii of a variable point moving on the parabola, $y^2 = 4ax$ is a parabola whose

(A) Latus rectum is half the latus rectum of the original parabola
(B) Vertex is $(a/2, 0)$
(C) Directrix is y-axis
(D) Focus has the co-ordinates $(a, 0)$

PR0163

11. P is a point on the parabola $y^2 = 4ax$ $(a > 0)$ whose vertex is A. PA is produced to meet the directrix in D and M is the foot of the perpendicular from P on the directrix. If a circle is described on MD as a diameter then it intersects the x-axis at a point whose co-ordinates are :

(A) $(-3a, 0)$
(B) $(-a, 0)$
(C) $(-2a, 0)$
(D) $(a, 0)$

PR0164

12. Let $y^2 = 4ax$ be a parabola and $x^2 + y^2 + 2\,bx = 0$ be a circle. If parabola and circle touch each other externally then :

(A) $a > 0, b > 0$
(B) $a > 0, b < 0$
(C) $a < 0, b > 0$
(D) $a < 0, b < 0$

PR0165

13. P is a point on the parabola $y^2 = 4x$ where abscissa and ordinate are equal. Equation of a circle passing through the focus and touching the parabola at P is :

(A) $x^2 + y^2 - 13x + 2y + 12 = 0$
(B) $x^2 + y^2 - 3x - 18y + 2 = 0$
(C) $x^2 + y^2 + 13x - 2y - 14 = 0$
(D) $x^2 + y^2 - x = 0$

PR0166

14. Subset of complete set of values of m for which a chord of slope m of the circle $x^2 + y^2 = 4$ touches parabola $y^2 = 4x$, can be

(A) $\left(-\infty, -\sqrt{\frac{\sqrt{2}-1}{2}}\right)$
(B) $(0\,,\,1/2)$
(C) $\left(\sqrt{\frac{\sqrt{2}-1}{2}}, \infty\right)$
(D) $(-1/2, 0)$

PR0167

15. Locus of the centre of the circle passing through the vertex and the mid-points of perpendicular chords from the vertex of the parabola $y^2 = 4ax$ is

(A) is a parabola with vertex $(-a, a)$
(B) is a parabola with latus rectum a
(C) is a parabola with vertex $(2a, 0)$
(D) is a parabola with latus rectum $\frac{a}{2}$

PR0168

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for question nos. 1 to 3

Tangents are drawn to the parabola $y^2 = 4x$ from the point $P(6, 5)$ to touch the parabola at Q and R. C_1 is a circle which touches the parabola at Q and C_2 is a circle which touches the parabola at R. Both the circles C_1 and C_2 pass through the focus of the parabola.

1. Area of the ΔPQR equals

(A) $\frac{1}{2}$ (B) 1 (C) 2 (D) $\frac{1}{4}$

PR0080

2. Radius of the circle C_2 is

(A) $5\sqrt{5}$ (B) $5\sqrt{10}$ (C) $10\sqrt{2}$ (D) $\sqrt{210}$

PR0080

3. The common chord of the circles C_1 and C_2 passes through the

(A) incentre of the ΔPQR (B) circumcenter of the ΔPQR

(C) centroid of the ΔPQR (D) orthocenter of the ΔPQR

PR0080

Paragraph for question nos. 4 to 6

Consider three lines y-axis, $y = 2$ and $\ell x + my = 1$ where (ℓ, m) lies on $y^2 = 4x$. Answer the following:

4. Locus of circum centre of triangle formed by given three lines is a parabola whose

(A) vertex is $\left(-2, \frac{3}{2}\right)$ (B) vertex is $\left(2, \frac{3}{2}\right)$ (C) L.R. = $\frac{1}{8}$ (D) L.R. = $\frac{1}{16}$

PR0169

5. Area of triangle formed by vertex and end points of latus rectum of parabola obtained in questions (4) is

(A) $\frac{1}{2^8}$ (unit)^2

(B) $\frac{1}{2^9}$ (unit)^2

(C) $\frac{1}{2^{10}}$ (unit)^2

(D) (Total number of subsets of a set containg 9 elements)$^{-1}$ (unit)^2 **PR0170**

6. Any point on the parabola obtained in question (4) can be represented as

(A) $\left(2+\frac{1}{32}t^2, \frac{3}{2}+\frac{t}{16}\right)$ (B) $\left(2+\frac{t^2}{32}, \frac{-3}{2}+\frac{1}{16}t^2\right)$

(C) $\left(-2+\frac{1}{32}t^2, \frac{3}{2}+\frac{t}{16}\right)$ (D) $\left(-2+\frac{1}{16}t^2, \frac{3}{2}+\frac{t}{5}\right)$ **PR0171**

Paragraph for Question 7 to 8

The normal at any point P to the parabola $y^2 = 8x$ makes an angle $\theta \in \left(0, \frac{\pi}{4}\right]$ with the positive direction of x-axis.

On the basis of above information, answer the following questions :

7. Maximum distance of the point P from origin is $2\sqrt{\lambda}$, then λ is - **PR0172**

8. Maximum focal distance of P is - **PR0173**

Matrix Match Type

9. Consider the parabola $y^2 = 12x$

List-I		**List-II**
(I) Tangent and normal at the extremities of the latus rectum intersect the x axis at T and G respectively. The coordinates of the middle point of T and G are **PR0081**	(P)	(0, 0)
(II) Variable chords of the parabola passing through a fixed point K on the axis, such that sum of the squares of the reciprocals of the two parts of the chords through K, is a constant. The coordinate of the point K are **PR0082**	(Q)	(3, 0)
(III) All variable chords of the parabola subtending a right angle at the origin are concurrent at the point **PR0083**	(R)	(6, 0)
(IV) AB and CD are the chords of the parabola which intersect at a point E on the axis. The radical axis of the two circles described on AB and CD as diameter always passes through **PR0084**	(S)	(12, 0)

(A) I $\to$ P; II $\to$ Q; III $\to$ R; IV $\to$ S
(B) I $\to$ Q; II $\to$ R; III $\to$ S; IV $\to$ P
(C) I $\to$ S; II $\to$ R; III $\to$ Q; IV $\to$ P
(D) I $\to$ R; II $\to$ Q; III $\to$ P; IV $\to$ S

10. AB is a chord of the parabola $y^2 = 4ax$ joining $A(at_1^2, 2at_1)$ and $B(at_2^2, 2at_2)$. Match the following :

Column - I	**Column - II**
(A) AB is a normal chord	(P) $t_2 = -t_1 + 2$
(B) AB is a focal chord	(Q) $t_2 = -\frac{4}{t_1}$
(C) AB subtends 90° at (0, 0)	(R) $t_2 = -\frac{1}{t_1}$
(D) AB is inclined at 45° to the axis of parabola	(S) $t_2 = -t_1 - \frac{2}{t_1}$

PR0174

EXERCISE (O-4)

Numerical Grid Type

1. If $(a^2, a-2)$ be a point interior to the region of the parabola $y^2 = 2x$ bounded by the chord joining the points $(2, 2)$ and $(8, -4)$, then the number of all possible integral values of a is :

PR0175

2. The number of integral values of a for which the point $(-2a, a + 1)$ will be an interior point of the smaller region bounded by the circle $x^2 + y^2 = 4$ and the parabola $y^2 = 4x$, is :

PR0176

3. A variable chord PQ of the parabola, $y^2 = 4x$ is drawn parallel to the line $y = x$. If the parameters of the points P & Q on the parabola be p & q respectively, then $(p + q)$ equal to

PR0177

4. The parabola whose axis is parallel to the y-axis and which passes through the points $(0, 4)$, $(1, 9)$ and $(-2, 6)$, also passes through $(2, \alpha)$ then the value of α is :

PR0178

5. The centre of the circle which passes through the focus of the parabola $x^2 = 4y$ & touches it at the point $(6, 9)$ is (α, β) then $|\alpha - \beta|$ is

PR0179

6. If ℓ is the distance between focus and directrix of the parabola $9x^2 - 24xy + 16y^2 - 20x - 15y - 60 = 0$ then ℓ is : **PR0180**

7. Points A, B & C lie on the parabola $y^2 = 4ax$. The tangents to the parabola at A, B & C, taken in pairs, intersect at points P, Q & R. the ratio of the areas of the triangles PQR & ABC is

PR0181

8. A normal is drawn to a parabola $y^2 = 4ax$ at any point other than the vertex and it cuts the parabola again at a point whose distance from the vertex is not less than ka find k

PR0182

9. If three normal are drawn through $(5c, 0)$ to $y^2 = 4x$ and two of which of perpendicular then the value of c is **PR0183**

10. Through the vertex O of the parabola $y^2 = 8x$, a perpendicular is drawn to any tangent meeting it at P & the parabola at Q, then the value of $\frac{OP.OQ}{10}$ is **PR0184**

EXERCISE (JM)

1. The radius of a circle, having minimum area, which touches the curve $y = 4 - x^2$ and the lines, $y = |x|$ is :- **[JEE-Main 2017]**

(1) $4(\sqrt{2}+1)$ (2) $2(\sqrt{2}+1)$ (3) $2(\sqrt{2}-1)$ (4) $4(\sqrt{2}-1)$

PR0127

2. Tangent and normal are drawn at P(16, 16) on the parabola $y^2 = 16x$, which intersect the axis of the parabola at A and B, respectively. If C is the centre of the circle through the points P, A and B and $\angle CPB = \theta$, then a value of $\tan\theta$ is - **[JEE-Main 2018]**

(1) 2 (2) 3 (3) $\frac{4}{3}$ (4) $\frac{1}{2}$ **PR0128**

3. If the tangent at (1, 7) to the curve $x^2 = y - 6$ touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ then the value of c is : **[JEE-Main 2018]**

(1) 185 (2) 85 (3) 95 (4) 195 **PR0129**

4. Let A(4,–4) and B(9,6) be points on the parabola, $y^2 = 4x$. Let C be chosen on the arc AOB of the parabola, where O is the origin, such that the area of ΔACB is maximum. Then, the area (in sq. units) of ΔACB, is: **[JEE (Main) 2019]**

(1) $31\frac{3}{4}$ (2) 32 (3) $30\frac{1}{2}$ (4) $31\frac{1}{4}$ **PR0130**

5. If the parabolas $y^2=4b(x-c)$ and $y^2=8ax$ have a common normal, then which one of the following is a valid choice for the ordered triad (a,b,c) **[JEE (Main)-2019]**

(1) (1, 1, 0) (2) $\left(\frac{1}{2},2,3\right)$ (3) $\left(\frac{1}{2},2,0\right)$ (4) (1, 1, 3) **PR0131**

6. If the tangent to the parabola $y^2 = x$ at a point (α, β), $(\beta > 0)$ is also a tangent to the ellipse, $x^2 + 2y^2 = 1$, then α is equal to : **[JEE (Main)-2019]**

(1) $2\sqrt{2}+1$ (2) $\sqrt{2}-1$ (3) $\sqrt{2}+1$ (4) $2\sqrt{2}-1$ **PR0132**

7. The area (in sq. units) of the smaller of the two circles that touch the parabola, $y^2 = 4x$ at the point (1, 2) and the x-axis is :- **[JEE (Main)-2019]**

(1) $4\pi(2-\sqrt{2})$ (2) $8\pi(3-2\sqrt{2})$ (3) $4\pi(3+\sqrt{2})$ (4) $8\pi(2-\sqrt{2})$

PR0133

8. The tangents to the curve $y=(x-2)^2-1$ at its points of intersection with the line $x-y=3$, intersect at the point : **[JEE (Main)-2019]**

(1) $\left(-\frac{5}{2},-1\right)$ (2) $\left(-\frac{5}{2},1\right)$ (3) $\left(\frac{5}{2},-1\right)$ (4) $\left(\frac{5}{2},1\right)$

PR0134

9. If $y = mx + 4$ is a tangent to both the parabolas, $y^2 = 4x$ and $x^2 = 2by$, then b is equal to :

[JEE (Main)-2020]

(1) 128 (2) –64 (3) –128 (4) –32

PR0136

10. Let a line $y = mx$ $(m > 0)$ intersect the parabola, $y^2 = x$ at a point P, other than the origin. Let the tangent to it at P meet the x-axis at the point Q. If area $(\Delta OPQ) = 4$ sq. units, then m is equal to.

[JEE (Main)-2020]

PR0137

11. The locus of a point which divides the line segment joining the point (0,–1) and a point on the parabola, $x^2 = 4y$, internally in the ratio 1 : 2, is- **[JEE (Main)-2020]**

(1) $9x^2 - 3y = 2$ (2) $9x^2 - 12y = 8$

(3) $x^2 - 3y = 2$ (4) $4x^2 - 3y = 2$ **PR0138**

12. If one end of a focal chord AB of the parabola $y^2 = 8x$ is at $A\left(\frac{1}{2},-2\right)$, then the equation of the tangent to it at B is : **[JEE (Main)-2020]**

(1) $2x + y - 24 = 0$ (2) $x - 2y + 8 = 0$

(3) $2x - y - 24 = 0$ (4) $x + 2y + 8 = 0$ **PR0139**

13. Let the tangent to the parabola $S : y^2 = 2x$ at the point P(2, 2) meet the x-axis at Q and normal at it meet the parabola S at the point R. Then the area (in sq. units) of the triangle PQR is equal to :

[JEE (Main)-2021]

(1) $\frac{25}{2}$ (2) $\frac{35}{2}$ (3) $\frac{15}{2}$ (4) 25

PR0185

14. Let $y = mx + c, m > 0$ be the focal chord of $y^2 = -64x$, which is tangent to $(x + 10)^2 + y^2 = 4$. Then, the value of $4\sqrt{2}$ (m + c) is equal to _______. **[JEE (Main)-2021]**

PR0186

15. A tangent and a normal are drawn at the point P(2, – 4) on the parabola $y^2 = 8x$, which meet the directrix of the parabola at the points A and B respectively. If Q(a, b) is a point such that AQBP is a square, then 2a + b is equal to : **[JEE (Main)-2021]**

(1) –16 (2) –18 (3) –12 (4) –20

PR0187

16. If $y = m_1x + c_1$ and $y = m_2x + c_2$, $m_1 \neq m_2$ are two common tangents of circle $x^2 + y^2 = 2$ and parabola $y^2 = x$, then the value of $8|m_1m_2|$ is equal to **[JEE (Main)-2022]**

(A) $3+4\sqrt{2}$ (B) $-5+6\sqrt{2}$ (C) $-4+3\sqrt{2}$ (D) $7+6\sqrt{2}$

PR0188

17. If the equation of the parabola, whose vertex is at (5, 4) and the directrix is $3x + y - 29 = 0$, is $x^2 + ay^2 + bxy + cx + dy + k = 0$ then $a + b + c + d + k$ is equal to **[JEE (Main)-2022]**

(A) 575 (B) –575 (C) 576 (D) –576

PR0189

18. Let P : $y^2 = 4ax$, $a > 0$ be a parabola with focus S. Let the tangents to the parabola P make an angle of $\frac{\pi}{4}$ with the line $y = 3x + 5$ touch the parabola P at A and B. Then the value of a for which A, B and S are collinear is : **[JEE (Main)-2022]**

(A) 8 only (B) 2 only (C) $\frac{1}{4}$ only (D) $a > 0$ only

PR0190

EXERCISE (JA)

Paragraph for Question 1 and 3

Let PQ be a focal chord of the parabolas $y^2 = 4ax$. The tangents to the parabola at P and Q meet at a point lying on the line $y = 2x + a$, $a > 0$.

1. If chord PQ subtends an angle θ at the vertex of $y^2 = 4ax$, then $\tan\theta =$

[JEE(Advanced) 2013, 3, (–1)]

(A) $\frac{2}{3}\sqrt{7}$ (B) $\frac{-2}{3}\sqrt{7}$ (C) $\frac{2}{3}\sqrt{5}$ (D) $\frac{-2}{3}\sqrt{5}$ **PR0146**

2. Length of chord PQ is **[JEE(Advanced) 2013, 3, (–1)]**

(A) 7a (B) 5a (C) 2a (D) 3a **PR0146**

3. The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the point P, Q and the parabola at the points R,S. Then the area of the quadrilateral PQRS is -

(A) 3 (B) 6 (C) 9 (D) 15 **PR0148**

[JEE(Advanced)-2014, 3(–1)]

Paragraph for Question 4 and 5

Let a,r,s,t be nonzero real numbers. Let $P(at^2, 2at)$, Q, $R(ar^2, 2ar)$ and $S(as^2, 2as)$ be distinct points on the parabola $y^2 = 4ax$. Suppose that PQ is the focal chord and lines QR and PK are parallel, where K is the point (2a,0).

4. The value of r is- **[JEE(Advanced)-2014, 3(–1)]**

(A) $-\frac{1}{t}$ (B) $\frac{t^2+1}{t}$ (C) $\frac{1}{t}$ (D) $\frac{t^2-1}{t}$ **PR0149**

5. If st = 1, then the tangent at P and the normal at S to the parabola meet at a point whose ordinate is-

[JEE(Advanced)-2014, 3(–1)]

(A) $\frac{\left(t^2+1\right)^2}{2t^3}$ (B) $\frac{a\left(t^2+1\right)^2}{2t^3}$ (C) $\frac{a\left(t^2+1\right)^2}{t^3}$ (D) $\frac{a\left(t^2+2\right)^2}{t^3}$

PR0149

6. If the normals of the parabola $y^2 = 4x$ drawn at the end points of its latus rectum are tangents to the circle $(x-3)^2 + (y+2)^2 = r^2$, then the value of r^2 is **[JEE 2015, 4M, –0M]**

PR0150

7. Let the curve C be the mirror image of the parabola $y^2 = 4x$ with respect to the line $x + y + 4 = 0$. If A and B are the points of intersection of C with the line $y = -5$, then the distance between A and B is

[JEE 2015, 4M, –0M]

PR0151

8. Let P and Q be distinct points on the parabola $y^2 = 2x$ such that a circle with PQ as diameter passes through the vertex O of the parabola. If P lies in the first quadrant and the area of the triangle ΔOPQ is $3\sqrt{2}$, then which of the following is(are) the coordinates of P ? **[JEE 2015, 4M, –2M]**

(A) $\left(4, 2\sqrt{2}\right)$ (B) $\left(9, 3\sqrt{2}\right)$ (C) $\left(\frac{1}{4}, \frac{1}{\sqrt{2}}\right)$ (D) $\left(1, \sqrt{2}\right)$ **PR0152**

9. The circle $C_1: x^2 + y^2 = 3$, with centre at O, intersects the parabola $x^2 = 2y$ at the point P in the first quadrant. Let the tangent to the circle C_1 at P touches other two circles C_2 and C_3 at R_2 and R_3, respectively. Suppose C_2 and C_3 have equal radii $2\sqrt{3}$ and centres Q_2 and Q_3, respectively. If Q_2 and Q_3 lie on the y-axis, then - **[JEE(Advanced)-2016, 4(–2)]**

(A) $Q_2Q_3 = 12$

(B) $R_2R_3 = 4\sqrt{6}$

(C) area of the triangle OR_2R_3 is $6\sqrt{2}$

(D) area of the triangle PQ_2Q_3 is $4\sqrt{2}$

PR0153

10. Let P be the point on the parabola $y^2 = 4x$ which is at the shortest distance from the center S of the circle $x^2 + y^2 - 4x - 16y + 64 = 0$. Let Q be the point on the circle dividing the line segment SP internally. Then -

(A) $SP = 2\sqrt{5}$

(B) $SQ : QP = \left(\sqrt{5}+1\right) : 2$

(C) the x-intercept of the normal to the parabola at P is 6

(D) the slope of the tangent to the circle at Q is $\frac{1}{2}$ **[JEE(Advanced)-2016, 4(–2)]**

PR0154

11. If a chord, which is not a tangent, of the parabola $y^2 = 16x$ has the equation $2x + y = p$, and midpoint (h, k), then which of the following is(are) possible value(s) of p, h and k ?

[JEE(Advanced)-2017, 4(–2)]

(A) p = 5, h = 4, k = –3

(B) p = –1, h = 1, k = –3

(C) p = –2, h = 2, k = –4

(D) p = 2, h = 3, k = –4 **PR0155**

12. Let E denote the parabola $y^2 = 8x$. Let P = (–2, 4), and let Q and Q' be two distinct points on E such that the lines PQ and PQ' are tangents to E. Let F be the focus of E. Then which of the following statements is (are) TRUE? **[JEE(Advanced)-2021]**

(A) The triangle PFQ is a right-angled triangle

(B) The triangle QPQ' is a right-angled triangle

(C) The distance between P and F is $5\sqrt{2}$

(D) F lies on the line joining Q and Q'

Question Stem for Questions Nos. 13 and 14

Question Stem

Consider the region $R = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \geq 0 \text{ and } y^2 \leq 4 - x\}$. Let F be the family of all circles that are contained in R and have centers on the x-axis. Let C be the circle that has largest radius among the circles in F. Let (α,β) be a point where the circle C meets the curve $y^2 = 4 - x$. **[JEE(Advanced)-2021]**

13. The radius of the circle C is _____

14. The value of α is _____

15. Consider the parabola $y^2 = 4x$. Let S be the focus of the parabola. A pair of tangents drawn to the parabola from the point P = (–2, 1) meet the parabola at P_1 and P_2. Let Q_1 and Q_2 be points on the lines SP_1 and SP_2 respectively such that PQ_1 is perpendicular to SP_1 and PQ_2 is perpendicular to SP_2. Then, which of the following is/are **TRUE** ?

[JEE(Advanced)-2022]

(A) $SQ_1 = 2$ (B) $Q_1Q_2 = \frac{3\sqrt{10}}{5}$ (C) $PQ_1 = 3$ (D) $SQ_2 = 1$

PR0191

ANSWER KEY

Do yourself - 1

1. Parabola

2. Vertex : $\left(-\frac{7}{2}, \frac{5}{2}\right)$, Axis : $y = \frac{5}{2}$, Focus : $\left(-\frac{17}{4}, \frac{5}{2}\right)$, Directrix : $x = -\frac{11}{4}$; LR = 3

3. $4x^2 + y^2 - 4xy + 8x + 46y - 71 = 0$; Axis : $2x - y = 3$; LR $= 4\sqrt{5}$ unit

4. $(3x + 4y - 4)^2 = \pm 20(4x - 3y + 7)$

5. D **6.** D **7.** $(-1,2)$; $y = 2$; 4 ; $(0,2)$ **8.** $\left(4, \frac{9}{2}\right)$; $x = 4$; 2; $(4, 4)$

9. $\left(a, \frac{a}{2}\right)$; $x = a$; $2a$; $(a, 0)$ **10.** $(1,2)$; $y = 2$; 4 ; $(0,2)$ **12.** $9y^2 = 4ax$

15. $4\sqrt{3}$

Do yourself - 2

1. $\left(-\infty, -\sqrt{\frac{8}{7}}\right) \cup \left(\sqrt{\frac{8}{7}}, \infty\right)$ **2.** $(-7, 8), (9, 8)$ **4.** $kb^2 = ac$

5. $8\sqrt{3}$ **7.** $y = -2x$; $y \pm 12 = m(x - 24)$

8. A **9.** C **10.** C

Do yourself - 3

1. $x - y + 3 = 0$, $(3, 6)$; $3x - 2y + 4 = 0$, $\left(\frac{4}{3}, 4\right)$

2. $2x - y + 2 = 0$, $(1, 4)$; $x + 2y + 16 = 0$, $(16, -16)$

4. $4y = x + 24$ **6.** $4y = x + 28$; $(28,14)$

7. $y + 2x + 1 = 0$; $\left(\frac{1}{2}, -2\right)$; $2y = x + 8$; $(8, 8)$ **8.** $4y = 9x + 4$; $4y = x + 36$.

11. $b^{\frac{1}{3}}y + a^{\frac{1}{3}}x + a^{\frac{2}{3}}b^{\frac{2}{3}} = 0$ **12.** $x = 0$ **13.** B **14.** A

15. $2x - y + 2 = 0$, $(1, 4)$; $x + 2y + 16 = 0$, $(16, -16)$

16. $3x - 2y + 4 = 0$; $x - y + 3 = 0$ **18.** 72 **20.** 3

Do yourself - 4

1. C **2.** 1 **3.** 48 **4.** (0, 0), (–4, 3) and (16, 8)

5. $4x + 3y = 34$ **6.** $x + y = 4a$, $x - y = 4a$ **7.** $\left(3a, 2\sqrt{3}a\right); \left(\frac{a}{3}, -\frac{2\sqrt{3}}{3}a\right)$

8. B **9.** B **10.** B **11.** C **12.** A **13.** C

14. B **15.** $y = -4x + 72$, $y = 3x - 33$ **17.** (a) $\frac{k-4}{h}$; (b) 2; (c) $2y - 3 = 0$

19. 512 **20.** $a^2 > 8b^2$

Do yourself - 5

1. $(y + 2)^2 = 28(x - 1)$ **2.** $\pi/2$ **3.** B

4. (i) $x = 2$ (ii) $y = -1$

5. (i) $[2x^2 - 3y^2 - xy + 13x + 3y + 18 = 0]$ (ii) $[2x^2 - y^2 + xy - 8x - 2y + 8 = 0]$

Do yourself - 6

1. $2x = y + 1$ **2.** (14, 12) **3.** $y^2 = 2a(x - a)$ **4.** $4x + 3y + 1 = 0$

6. C **7.** (4 , 0) ; $y^2 = 2a(x - 4a)$ **8.** $x - y = 1$; $8\sqrt{2}$ sq. units

9. 32 **10.** 25/2

Do yourself - 7

1. 1 : 1 : 1 **2.** $y^2 - 4x + 2 = 0$ **6.** (a) 4, (b) (2,–1), (c) 8 sq. units

7. 11

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	A	D	A	D	B	D	C	D	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	C	A	C	B	B	C	C	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	D	D	C	B	D	C	B	D

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	A,B	A,B,C	A,D	A,B,C,D	A,B,C,D	A,B,D	A,C	A,B	A,B,C,D
Que.	11	12	13	14	15					
Ans.	A,D	A,D	A,D	A,C	B,C					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	A	B	C	A,C	B,D	C	5.00	4.00	B
Q.10	A	B	C	D					
	S	R	Q	P					

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	0	2	18	23	0.50	3.00	9.76 to 9.80	0.60	1.60

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7
Ans.	Bonus, correct answer is $\frac{1}{2}\left(\sqrt{17}-\sqrt{2}\right)$	1	3	4	1,2,3,4	3	2

Que.	8	9	10	11	12	13	14	15	16	17
Ans.	3	3	0.50	2	2	1	34	1	C	D
Que.	18									
Ans.	D									

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	D	D	B	2	4	A,D	A,B,C	A,C,D
Que.	11	12	13	14	15					
Ans.	D	A,B,D	1.50	2.00	B,C,D					

EXERCISE (O-1)

Straight Objective Type

1. Let 'E' be the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ & 'C' be the circle $x^2 + y^2 = 9$. Let P & Q be the points (1 , 2) and (2,1) respectively. Then :

(A) Q lies inside C but outside E
(B) Q lies outside both C & E
(C) P lies inside both C & E
(D) P lies inside C but outside E. **EL0001**

2. The eccentricity of the ellipse $(x-3)^2+(y-4)^2=\frac{y^2}{9}$ is

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{3\sqrt{2}}$ (D) $\frac{1}{\sqrt{3}}$ **EL0002**

3. The equation, $2x^2 + 3y^2 - 8x - 18y + 35 = K$ represents

(A) no locus if $K > 0$
(B) an ellipse if $K < 0$
(C) a point if $K = 0$
(D) a hyperbola if $K > 0$ **EL0003**

4. If $P = (x,y)$, $F_1 = (3,0)$, $F_2 = (-3,0)$ and $16x^2 + 25y^2 = 400$, then $PF_1 + PF_2$ equals

(A) 8 (B) 6 (C) 10 (D) 12 **EL0108**

5. If the ellipse $\frac{(x-h)^2}{M}+\frac{(y-k)^2}{N}=1$ has major axis on the line $y = 2$, minor axis on the line $x = -1$, major axis has length 10 and minor axis has length 4. The number h,k,M,N (in this order only) are -

(A) –1,2,5,2 (B) –1,2,10,4 (C) 1,–2,25,4 (D) –1,2,25,4 **EL0004**

6. An ellipse has OB as a semi-minor axis. F and F' are its foci and the angle FBF' is a right angle. Then, the eccentricity of the ellipse is

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$ **EL0109**

7. The y-axis is the directrix of the ellipse with eccentricity $e = 1/2$ and the corresponding focus is at (3,0), equation to its auxiliary circle is

(A) $x^2 + y^2 - 8x + 12 = 0$
(B) $x^2 + y^2 - 8x - 12 = 0$
(C) $x^2 + y^2 - 8x + 9 = 0$
(D) $x^2 + y^2 = 4$ **EL0005**

8. Imagine that you have two thumbtacks placed at two distinct points, A and B. If the ends of a fixed length of string are fastened to the thumbtacks and the string is drawn taut with a pencil, the path traced by the pencil will be a curve called ellipse. The best way to maximise the area surrounded by the curve with a fixed length of string occurs when

I the two points A and B have the maximum distance between them.

II two points A and B coincide.

III A and B are placed vertically.

IV The area is always same regardless of the location of A and B .

(A) I (B) II (C) III (D) IV **EL0006**

9. The latus rectum of a conic section is the width of the function through the focus. The positive difference between the length of the latus rectum of $3y = x^2 + 4x - 9$ and $x^2 + 4y^2 - 6x + 16y = 24$ is-

(A) $\frac{1}{2}$ (B) 2 (C) $\frac{3}{2}$ (D) $\frac{5}{2}$ **EL0007**

10. Let S(5,12) and S'(–12,5) are the foci of an ellipse passing through the origin. The eccentricity of ellipse equals -

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{2}{3}$ **EL0008**

11. A circle has the same centre as an ellipse & passes through the foci F_1 & F_2 of the ellipse, such that the two curves intersect in 4 points. Let 'P' be any one of their point of intersection. If the major axis of the ellipse is 17 & the area of the triangle PF_1F_2 is 30, then the distance between the foci is :

(A) 11 (B) 12 (C)13 (D) none **EL0009**

12. An ellipse is inscribed in a circle and a point within the circle is chosen at random. If the probability that this point lies outside the ellipse is 2/3 then the eccentricity of the ellipse is :

(A) $\frac{2\sqrt{2}}{3}$ (B) $\frac{\sqrt{5}}{3}$ (C) $\frac{8}{9}$ (D) $\frac{2}{3}$ **EL0010**

13. (a) Which of the following is an equation of the ellipse with centre (–2,1), major axis running from (–2,6) to (–2,–4) and focus at (–2,5) ?

(A) $\frac{(x-2)^2}{25}+\frac{(y+1)^2}{16}=1$ (B) $\frac{(x+2)^2}{25}+\frac{(y-1)^2}{9}=1$

(C) $\frac{(x-2)^2}{9}+\frac{(y+1)^2}{25}=1$ (D) $\frac{(x+2)^2}{9}+\frac{(y-1)^2}{25}=1$

(b) Which of the following statement(s) is/are correct for the ellipse of 11(a) ?

(A) auxiliary circle is $(x + 2)^2 + (y - 1)^2 = 25$ (B) director circle is $(x + 2)^2 + (y - 1)^2 = 34$

(C) Latus rectum $=\frac{18}{5}$ (D) eccentricity $=\frac{4}{5}$ **EL0011**

14. $x-2y+4=0$ is a common tangent to $y^2=4x$ & $\frac{x^2}{4}+\frac{y^2}{b^2}=1$. Then the value of b and the other common tangent are given by :

(A) $b=\sqrt{3}$; $x+2y+4=0$ (B) $b=3$; $x+2y+4=0$

(C) $b=\sqrt{3}$; $x+2y-4=0$ (D) $b=\sqrt{3}$; $x-2y-4=0$ **EL0012**

15. Consider the particle travelling clockwise on the elliptical path $\frac{x^2}{100}+\frac{y^2}{25}=1$. The particle leaves the orbit at the point (–8, 3) and travels in a straight line tangent to the ellipse. At what point will the particle cross the y-axis ?

(A) $\left(0,\frac{25}{3}\right)$ (B) $\left(0,\frac{23}{3}\right)$ (C) (0, 9) (D) $\left(0,\frac{26}{3}\right)$ **EL0013**

16. The number of values of c such that the straight line $y=4x+c$ touches the curve $\frac{x^2}{4}+y^2=1$ is

(A) 0 (B) 2 (C) 1 (D) ∞ **EL0110**

17. If α & β are the eccentric angles of the extremities of a focal chord of an standard ellipse, then the eccentricity of the ellipse is :

(A) $\frac{\cos\alpha+\cos\beta}{\cos(\alpha+\beta)}$ (B) $\frac{\sin\alpha-\sin\beta}{\sin(\alpha-\beta)}$

(C) $\frac{\cos\alpha-\cos\beta}{\cos(\alpha-\beta)}$ (D) $\frac{\sin\alpha+\sin\beta}{\sin(\alpha+\beta)}$ **EL0025**

18. If P is any point on ellipse with foci S_1 & S_2 and eccentricity is $\frac{1}{2}$ such that

$\angle PS_1S_2=\alpha$, $\angle PS_2S_1=\beta$, $\angle S_1PS_2=\gamma$, then $\cot\frac{\alpha}{2}$, $\cot\frac{\gamma}{2}$, $\cot\frac{\beta}{2}$ are in

(A) A.P. (B) G.P.

(C) H.P. (D) ***NOT*** A.P., G.P. or H.P. **EL0026**

19. Equation of the common tangent to the ellipses, $\frac{x^2}{a^2+b^2}+\frac{y^2}{b^2}=1$ and $\frac{x^2}{a^2}+\frac{y^2}{a^2+b^2}=1$ is -

(A) $ay=bx+\sqrt{a^4-a^2b^2+b^4}$ (B) $by=ax-\sqrt{a^4+a^2b^2+b^4}$

(C) $ay=bx-\sqrt{a^4+a^2b^2+b^4}$ (D) $by=ax+\sqrt{a^4-a^2b^2+b^4}$ **EL0027**

20. The normal at a variable point P on an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ of eccentricity e meets the axes of the ellipse in Q and R then the locus of the mid-point of QR is a conic with an eccentricity e' such that :

(A) e' is independent of e (B) $e' = 1$

(C) $e' = e$ (D) $e' = 1/e$ **EL0028**

21. The area of the rectangle formed by the perpendiculars from the centre of the standard ellipse to the tangent and normal at its point whose eccentric angle is $\pi/4$, is :

(A) $\frac{(a^2-b^2)ab}{a^2+b^2}$ (B) $\frac{(a^2-b^2)}{(a^2+b^2)ab}$ (C) $\frac{(a^2-b^2)}{ab(a^2+b^2)}$ (D) $\frac{a^2+b^2}{(a^2-b^2)ab}$ **EL0029**

22. The locus of the middle point of chords of an ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ passing through P(0, 5) is another ellipse E. The coordinates of the foci of the ellipse E, is

(A) $\left(0, \frac{3}{5}\right)$ and $\left(0, \frac{-3}{5}\right)$ (B) (0, – 4) and (0, 1)

(C) (0, 4) and (0, 1) (D) $\left(0, \frac{11}{2}\right)$ and $\left(0, \frac{-1}{2}\right)$ **EL0030**

23. **Statement-1 :** The ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $\frac{x^2}{9}+\frac{y^2}{16}=1$ are congruent.

and

Statement-2 : The ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $\frac{x^2}{9}+\frac{y^2}{16}=1$ have the same eccentricity.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True. **EL0018**

24. Tangents are drawn from the point $P\left(-\sqrt{3},\sqrt{2}\right)$ to an ellipse $4x^2 + y^2 = 4$.

Statement-1 : The tangents are mutually perpendicular.

and

Statement-2 : The locus of the points from which mutually perpendicular tangents can be drawn to given ellipse is $x^2 + y^2 = 5$.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True. **EL0019**

25. Suppose x and y are real numbers such that $x^2 + 9y^2 - 4x + 6y + 4 = 0$ then maximum value of $(4x - 9y)$ is **EL0041**

(A) 10 (B) 16 (C) 32 (D) 22

26. A tangent having slope $-\frac{4}{3}$ to the ellipse $\frac{x^2}{18}+\frac{y^2}{32}=1$, intersects the axis of x & y in points A & B respectively. If O is the origin, then area of triangle OAB is . **EL0049**

(A) 12 sq. units (B) 36 sq. units (C) 24 sq. units (D) 30 sq. units

27. Tangents drawn from the point P(2,3) to the circle $x^2 + y^2 - 8x + 6y + 1 = 0$ touch the circle at the points A and B. The circumcircle of the ΔPAB cuts the director circle of ellipse $\frac{(x+5)^2}{9}+\frac{(y-3)^2}{b^2}=1$ orthogonally, then the value of b^2 is **EL0050**

(A) 44 (B) 24 (C) 64 (D) 54

28. An ellipse has foci at $F_1(9, 20)$ and $F_2(49, 55)$ in the xy-plane and is tangent to the x-axis, then the length of its major axis is **EL0059**

(A) 85 (B) 170 (C) 70 (D) 140

29. The equation of the largest circle with centre (1, 0) that can be inscribed in the ellipse $x^2 + 4y^2 = 16$ is **EL0055**

(A) $(x + 1)^2 + y^2 = \frac{11}{3}$ (B) $(x - 1)^2 + y^2 = \frac{11}{3}$

(C) $(x - 1)^2 + y^2 = \frac{13}{3}$ (D) $(x + 1)^2 + y^2 = \frac{13}{3}$

30. Point 'O' is the centre of the ellipse with major axis AB & minor axis CD. Point F is one focus of the ellipse. If OF = 6 & the diameter of the inscribed circle of triangle OCF is 2, then the product (AB) (CD) is **EL0042**

(A) 55 (B) 45 (C) 65 (D) 75

EXERCISE (O-2)

Multiple Correct Answer Type

1. The equation of the standard ellipse with respect to coordinate axes whose minor axis is equal to the distance between its foci and whose LR = 10, will be-

(A) $2x^2 + y^2 = 100$
(B) $x^2 + 2y^2 = 100$
(C) $2x^2 + 3y^2 = 80$
(D) none of these

EL0115

2. Consider the ellipse $\frac{x^2}{4}+\frac{y^2}{b^2}=1$ $(b^2 < 4)$ and whose eccentricity is 'e'. If the normal at the end of latus-rectum passes through one extremity of minor axis, then which of the following statement(s) is/are true ?

(A) Equation of director circle of ellipse is $x^2 + y^2 = 10 - 2\sqrt{5}$

(B) Length of latus rectum of the ellipse is equal to b^2

(C) The value of $e^2 + e$ is 1

(D) Area of region between director and auxiliary circle of the ellipse is $2\pi\left(3-\sqrt{5}\right)$

EL0116

3. Let $A(\alpha)$ and $B(\beta)$ be the extremeties of a chord of an ellipse . If the slope of AB is equal to the slope of the tangent at a point $C(\theta)$ on the ellipse, then the value of θ, is

(A) $\frac{\alpha+\beta}{2}$
(B) $\frac{\alpha-\beta}{2}$
(C) $\frac{\alpha+\beta}{2}+\pi$
(D) $\frac{\alpha-\beta}{2}-\pi$

EL0117

4. Let F_1, F_2 be two foci of the ellipse and PT and PN be the tangent and the normal respectively to the ellipse at point P then

(A) PN bisects $\angle F_1 PF_2$
(B) PT bisects $\angle F_1 PF_2$
(C) PT bisects angle $(180° - \angle F_1 PF_2)$
(D) None of these

EL0118

5. If ℓ_1 be the equation of the common tangent in 1st quadrant to the circle $x^2 + y^2 = 16$ and ellipse $\frac{x^2}{25}+\frac{y^2}{4}=1$ and λ_1 be the length of the intercept of the common tangent between the coordinate axes then

(A) $\lambda_1 = \frac{14}{\sqrt{3}}$
(B) Equation of ℓ_1 is $2x+\sqrt{3}y=4\sqrt{7}$
(C) $\lambda_1 = \frac{4}{\sqrt{3}}$
(D) Equation of ℓ_1 is $x+\sqrt{3}y=4\sqrt{7}$

EL0119

6. If latus rectum of an ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ $\{0<b<4\}$, subtend angle 2θ at farthest vertex such that $\operatorname{cosec}\theta = \sqrt{5}$, then -

(A) $e=\frac{1}{2}$

(B) no such ellipse exist

(C) $b=2\sqrt{3}$

(D) area of Δ formed by LR and nearest vertex is 6 sq. units **EL0120**

7. If $x - 2y + k = 0$ is a common tangent to $y^2 = 4x$ & $\frac{x^2}{a^2}+\frac{y^2}{3}=1\left(a>\sqrt{3}\right)$, then the value of a, k and other common tangent are given by -

(A) a = 2 (B) a = 3 (C) $x + 2y + 4 = 0$ (D) k = 4 **EL0121**

8. Equation of the ellipse whose axes are the axes of coordinates and which passes through the point (–3, 1) and has eccentricity $\sqrt{2/5}$ is :-

(A) $3x^2 + 5y^2 - 15 = 0$ (B) $5x^2 + 3y^2 - 32 = 0$

(C) $3x^2 + 5y^2 - 32 = 0$ (D) $5x^2 + 3y^2 - 48 = 0$ **EL0076**

9. If a and c are positive real number and the ellipse $\frac{x^2}{4c^2}+\frac{y^2}{c^2}=1$ has four distinct points in common with the circle $x^2 + y^2 = 9a^2$, then

(A) $6ac + 9a^2 - 2c^2 > 0$ (B) $6ac + 9a^2 - 2c^2 < 0$

(C) $9ac - 9a^2 - 2c^2 < 0$ (D) $9ac - 9a^2 - 2c^2 > 0$ **EL0080**

10. Let the equations of two ellipses be $E_1 : \frac{x^2}{3}+\frac{y^2}{2}=1$ and $E_2 : \frac{x^2}{16}+\frac{y^2}{b^2}=1$. If the product of their eccentricities is $\frac{1}{2}$, then the length of the minor axis of ellipse E_2 can be :-

(A) 9 (B) 8 (C) 2 (D) 4 **EL0082**

11. Consider the ellipse $\frac{x^2}{\tan^2\alpha}+\frac{y^2}{\sec^2\alpha}=1$ where $\alpha \in (0,\pi/2)$.

Which of the following quantities would vary as α varies ?

(A) degree of flatness (B) ordinate of the vertex

(C) coordinates of the foci (D) length of the latus rectum **EL0014**

12. The equation of the common tangents of the parabola $y^2 = 4x$ and an ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ are -

(A) $x - 2y + 4 = 0$ (B) $x + 2y + 4 = 0$ (C) $2x - y + 1 = 0$ (D) $2x + y + 1 = 0$

EL0015

13. Consider an ellipse $\frac{x^2}{a}+\frac{y^2}{b}=1$. If A_1 and A_2 denotes area of ellipse and its director circle respectively, then -

(A) $A_2 \geq 2A_1$ (B) $A_2 < 2A_1$

(C) if $a = 4b$, then $A_1 = \frac{2}{5}A_2$ (D) if $a = 4b$, then $A_2 = \frac{2}{5}A_1$ **EL0016**

14. If length of perpendicular drawn from origin to any normal of the ellipse $\frac{x^2}{16}+\frac{y^2}{25}=1$ is ℓ, then ℓ cannot be -

(A) 4 (B) $\frac{5}{2}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$ **EL0017**

15. Extremities of the latus rectum of the ellipses $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ $(a > b)$ having a given major axis 2a lies on-

(A) $x^2 = a(a - y)$ (B) $x^2 = a(a + y)$ (C) $y^2 = a(a + x)$ (D) $y^2 = a(a - x)$ **EL0031**

16. If a number of ellipse (whose axes are x & y axes) be described having the same major axis 2a but a variable minor axis then the tangents at the ends of their latus rectum pass through fixed points which can be -

(A) (0,a) (B) (0,0) (C) (0,–a) (D) (a,a) **EL0032**

17. Tangents are drawn from any point on the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ to the circle $x^2 + y^2 = 1$ and respective chord of contact always touches a conic 'C', then -

(A) minimum distance between 'C' & ellipse is $\frac{3}{2}$

(B) maximum distance between 'C' & ellipse is $\frac{10}{3}$

(C) eccentricity of 'C' is $\frac{\sqrt{5}}{3}$

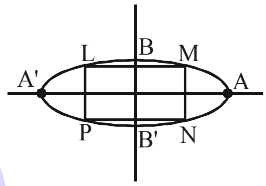
(D) product of eccentricity of 'C' & ellipse is 1 **EL0033**

18. Two lines are drawn from point $P(\alpha, \beta)$ which touches $y^2 = 8x$ at A, B and touches $\frac{x^2}{4}+\frac{y^2}{6}=1$ at C, D, then -

(A) $\alpha + \beta = -4$ (B) $\alpha\beta = 4$

(C) Area of triangle PAB is $128\sqrt{2}$ (D) Area of triangle PAB is $32\sqrt{2}$ **EL0034**

19. Let LMNP be a rectangle (which is not a square) inscribed in the ellipse as shown in the figure. Let 'λ' denotes the number of ways in which we can select four points out of A,A',B,B', L,M,P and N such that normal at these four points are concurrent, then 'λ' is less than or equal to-



(A) 5 (B) 6 (C) 7 (D) 4 **EL0035**

20. Let P(r, s), (r < 0) is a variable point on ellipse $C_1 : \frac{x^2}{16}+\frac{y^2}{9}=1$. PA and PB are two tangents of parabola $C_2 : y^2 - 4x = 0$, where A and B are point of contacts, then -

(A) number of common tangents of curves C_1 and C_2 is 4.

(B) minimum integral value of r + s is –5

(C) mid point of chord AB lie on curve $\frac{y^2}{9}+\frac{(y^2-2x)^2}{64}=1$

(D) mid point of chord AB lie on curve $\frac{x^2}{9}+\frac{(y^2-2x)^2}{64}=1$ **EL0036**

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for question nos. 1 to 3

Consider the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ and the parabola $y^2 = 2x$. They intersect at P and Q in the first and fourth quadrants respectively. Tangents to the ellipse at P and Q intersect the x-axis at R and tangents to the parabola at P and Q intersect the x-axis at S.

1. The ratio of the areas of the triangles PQS and PQR, is

(A) 1 : 3 (B) 1 : 2 (C) 2 : 3 (D) 3 : 4 **EL0020**

2. The area of quadrilateral PRQS, is

(A) $\frac{3\sqrt{15}}{2}$ (B) $\frac{15\sqrt{3}}{2}$ (C) $\frac{5\sqrt{3}}{2}$ (D) $\frac{5\sqrt{15}}{2}$ **EL0020**

3. The equation of circle touching the parabola at upper end of its latus rectum and passing through its vertex, is

(A) $2x^2 + 2y^2 - x - 2y = 0$ (B) $2x^2 + 2y^2 + 4x - \frac{9}{2}y = 0$

(C) $2x^2 + 2y^2 + x - 3y = 0$ (D) $2x^2 + 2y^2 - 7x + y = 0$ **EL0020**

Paragraph for question nos. 4 to 6

Tangents are drawn from the point P(3,4) to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ touching the ellipse at points A and B.

4. The coordinates of A and B are

(A) (3,0) and (0,2) (B) $\left(-\frac{8}{5},\frac{2\sqrt{261}}{15}\right)$ and $\left(-\frac{9}{5},\frac{8}{5}\right)$

(C) $\left(-\frac{8}{5},\frac{2\sqrt{161}}{15}\right)$ and (0,2) (D) (3,0) and $\left(-\frac{9}{5},\frac{8}{5}\right)$ **EL0096**

5. The orthocenter of the triangle PAB is

(A) $\left(5,\frac{8}{7}\right)$ (B) $\left(\frac{7}{5},\frac{25}{8}\right)$ (C) $\left(\frac{11}{5},\frac{8}{5}\right)$ (D) $\left(\frac{8}{25},\frac{7}{5}\right)$ **EL0096**

6. The equation of the locus of the point whose distances from the point P and the line AB are equal, is -

(A) $9x^2 + y^2 - 6xy - 54x - 62y + 241 = 0$ (B) $x^2 + 9y^2 + 6xy - 54x + 62y - 241 = 0$

(C) $9x^2 + 9y^2 - 6xy - 54x - 62y - 241 = 0$ (D) $x^2 + y^2 - 2xy + 27x + 31y - 120 = 0$

EL0096

Paragraph for question nos. 7 to 8

Let the two foci of an ellipse be (– 1, 0) and (3, 4) and the foot of perpendicular from the focus (3, 4) upon a tangent to the ellipse be (4, 6).

7. The foot of perpendicular from the focus (–1, 0) upon the same tangent to the ellipse is (a, b) then 5(a + b) equal to **EL0037**

8. The length of semi-minor axis of the ellipse is $\sqrt{\lambda}$ then $\frac{\lambda}{2}$ equal to **EL0037**

Matrix Match Type

9. Consider an ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ with centre C and a point P on it with eccentric angle $\frac{\pi}{4}$. Normal drawn at P intersects the major and minor axes in A and B respectively. N_1 and N_2 are the feet of the perpendiculars from the foci S_1 and S_2 respectively on the tangent at P and N is the foot of the perpendicular from the centre of the ellipse on the normal at P. Tangent at P intersects the axis of x at T.

	List-I		**List-II**
(I)	(CA)(CT) is equal to	(P)	9
(II)	(PN)(PB) is equal to	(Q)	16
(III)	$(S_1N_1)(S_2N_2)$ is equal to	(R)	17
(IV)	$(S_1P)(S_2P)$ is equal to	(S)	25
		(T)	23

EL0074

Which of the following is correct option :-

(A) (I) → (Q); (II) → (S); (III) → (P); (IV) → (R)

(B) (I) → (R); (II) → (T); (III) → (S); (IV) → (P)

(C) (I) → (S); (II) → (T); (III) → (Q); (IV) → (P)

(D) (I) → (Q); (II) → (S); (III) → (T); (IV) → (P)

10.

	Column-I		**Column-II**
(A)	The eccentricity of the ellipse which meets the straight line $2x - 3y = 6$ on the X-axis and the straight line $4x + 5y = 20$ on the Y-axis and whose principal axes lie along the coordinate axes, is **EL0021**	(P)	$\frac{1}{2}$
(B)	A bar of length 20 units moves with its ends on two fixed straight lines at right angles. A point P marked on the bar at a distance of 8 units from one end describes a conic whose eccentricity is **EL0022**	(Q)	$\frac{1}{\sqrt{2}}$
(C)	If one extremity of the minor axis of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and the foci form an equilateral triangle, then its eccentricity, is **EL0023**	(R)	$\frac{\sqrt{5}}{3}$
(D)	There are exactly two points on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ whose distance from the centre of the ellipse are greatest and equal to $\sqrt{\frac{a^2+2b^2}{2}}$. Eccentricity of this ellipse is equal to **EL0024**	(S)	$\frac{\sqrt{7}}{4}$

EXERCISE (O-4)

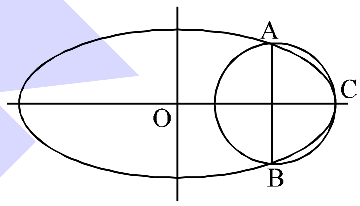
Numerical Grid Type

1. Equations of the line with equal intercepts on the axes & which touch the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ is ax + by + c = 0 then value of $\frac{a^2+b^2}{c^2}$ is **EL0048**

2. Let P_i and P_i' be the feet of the perpendiculars drawn from foci S,S' on a tangent T_i to an ellipse whose length of semi-major axis is 20. If $\sum_{i=1}^{10}(SP_i)(S'P_1')=2560$, then the value of 100e is (where e eccentricity) **EL0056**

3. Rectangle ABCD has area 200. An ellipse with area 200π passes through A and C and has foci at B and D. Find the perimeter of the rectangle. **EL0061**

4. A circle intersects an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ precisely at three points A, B, C as shown in the figure. AB is a diameter of the circle and is perpendicular to the major axis of the ellipse. If the eccentricity of the ellipse is 4/5, length of the diameter AB is λa, then find λ.



EL0046

5. Eccentricity of an ellipse in which distance between their foci is 10 and that of focus and corresponding directrix is 15 is e, then 4e is equal to **EL0122**

6. An ellipse passes through the points (–3, 1) & (2, –2) and its principal axes are along the coordinate axes in order. The value of 10 × (length of semi-minor axis)^2 is equal to **EL0123**

7. The tangent and normal to the ellipse $x^2 + 4y^2 = 4$ at a point P(θ) on it meet the major axis in Q and R respectively. If QR = 2 and the eccentric angle θ of P is given by $\cos\theta=\pm\left(\frac{a}{b}\right)$, where a & b are relative prime, then value of $\frac{2b}{a}$ is equal to **EL0124**

8. If x, y ∈ R, satisfies the equation $\frac{(x-4)^2}{4}+\frac{y^2}{9}=1$, then the difference between the largest and the smallest value of the expression $\frac{x^2}{4}+\frac{y^2}{9}$ is … **EL0125**

9. If $a(x^2 + y^2 + 2y + 1)=(x - 2y + 3)^2$ is an ellipse and a ∈ (b, ∞), then the value of b is **EL0126**

10. S_1 and S_2 are the foci of an ellipse of major axis of length 10 units, and P is any point on the ellipse such that the perimeter of triangle PS_1S_2 is 15. Then the eccentricity of the ellipse is **EL0127**

EXERCISE (JM)

1. The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and having centre at (0, 3) is : **[JEE (Main)-2013]**

(1) $x^2 + y^2 - 6y - 7 = 0$ (2) $x^2 + y^2 - 6y + 7 = 0$

(3) $x^2 + y^2 - 6y - 5 = 0$ (4) $x^2 + y^2 - 6y + 5 = 0$ **EL0079**

2. Equation of the line passing through the points of intersection of the parabola $x^2 = 8y$ and the ellipse $\frac{x^2}{3} + y^2 = 1$ is - **[JEE-Main (On line)-2013]**

(1) $y + 3 = 0$ (2) $3y + 1 = 0$ (3) $3y - 1 = 0$ (4) $y - 3 = 0$

EL0081

3. If the curves $\frac{x^2}{\alpha} + \frac{y^2}{4} = 1$ and $y^3 = 16x$ intersect at right angles, then a value of α is :

[JEE-Main (On line)-2013]

(1) $\frac{4}{3}$ (2) $\frac{3}{4}$ (3) $\frac{1}{2}$ (4) 2 **EL0083**

4. A point on the ellipse, $4x^2 + 9y^2 = 36$, where the normal is parallel to the line, $4x - 2y - 5 = 0$, is:

[JEE-Main (On line)-2013]

(1) $\left(\frac{8}{5},-\frac{9}{5}\right)$ (2) $\left(-\frac{9}{5},\frac{8}{5}\right)$ (3) $\left(\frac{8}{5},\frac{9}{5}\right)$ (4) $\left(\frac{9}{5},\frac{8}{5}\right)$ **EL0084**

5. The locus of the foot of perpendicular drawn from the centre of the ellipse $x^2 + 3y^2 = 6$ on any tangent to it is : **[JEE(Main)-2014]**

(1) $(x^2 - y^2)^2 = 6x^2 + 2y^2$ (2) $(x^2 - y^2)^2 = 6x^2 - 2y^2$

(3) $(x^2 + y^2)^2 = 6x^2 + 2y^2$ (4) $(x^2 + y^2)^2 = 6x^2 - 2y^2$ **EL0085**

6. The area (in sq. units) of the quadrilateral formed by the tangents at the end points of the latera recta to the ellipse $\frac{x^2}{9} + \frac{y^2}{5} = 1$ is : **[JEE (Main)-2015]**

(1) $\frac{27}{2}$ (2) 27 (3) $\frac{27}{4}$ (4) 18 **EL0086**

7. The eccentricity of an ellipse whose centre is at the origin is $\frac{1}{2}$. If one of its directices is $x = -4$, then the equation of the normal to it at $\left(1,\frac{3}{2}\right)$ is :- **[JEE(Main) 2017]**

(1) $x + 2y = 4$ (2) $2y - x = 2$ (3) $4x - 2y = 1$ (4) $4x + 2y = 7$

EL0087

8. If tangents are drawn to the ellipse $x^2 + 2y^2 = 2$ at all points on the ellipse other than its four vertices then the mid points of the tangents intercepted betwen the coordinate axes lie on the curve : **[JEE (Main)-2019]**

(1) $\frac{x^2}{2}+\frac{y^2}{4}=1$ (2) $\frac{x^2}{4}+\frac{y^2}{2}=1$ (3) $\frac{1}{2x^2}+\frac{1}{4y^2}=1$ (4) $\frac{1}{4x^2}+\frac{1}{2y^2}=1$

EL0088

9. Let S and S' be the foci of the ellipse and B be any one of the extremities of its minor axis. If ΔS'BS is a right angled triangle with right angle at B and area (ΔS'BS) = 8 sq. units, then the length of a latus rectum of the ellipse is : **[JEE (Main)-2019]**

(1) $2\sqrt{2}$ (2) 2 (3) 4 (4) $4\sqrt{2}$ **EL0089**

10. If the line x – 2y = 12 is tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at the point $\left(3,\frac{-9}{2}\right)$, then the length of the latus recturm of the ellipse is : **[JEE (Main)-2019]**

(1) 9 (2) $8\sqrt{3}$ (3) $12\sqrt{2}$ (4) 5 **EL0090**

11. If the normal to the ellipse $3x^2 + 4y^2 = 12$ at a point P on it is parallel to the line, 2x + y = 4 and the tangent to the ellipse at P passes through Q(4, 4) then PQ is equal to : **[JEE (Main)-2019]**

(1) $\frac{\sqrt{221}}{2}$ (2) $\frac{\sqrt{157}}{2}$ (3) $\frac{\sqrt{61}}{2}$ (4) $\frac{5\sqrt{5}}{2}$ **EL0091**

12. If $3x + 4y = 12\sqrt{2}$ is a tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{9}=1$ for some $a \in R$, then the distance between the foci of the ellipse is : **[JEE (Main)-2020]**

(1) 4 (2) $2\sqrt{7}$ (3) $2\sqrt{5}$ (4) $2\sqrt{2}$ **EL0092**

13. If the distance between the foci of an ellipse is 6 and the distance between its directrices is 12, then the length of its latus rectum is : **[JEE (Main)-2020]**

(1) $\sqrt{3}$ (2) $2\sqrt{3}$ (3) $3\sqrt{2}$ (4) $\frac{3}{\sqrt{2}}$ **EL0093**

14. Let the line y = mx and the ellipse $2x^2 + y^2 = 1$ intersect at a ponit P in the first quadrant. If the normal to this ellipse at P meets the co-ordinate axes at $\left(-\frac{1}{3\sqrt{2}},0\right)$ and $(0,\beta)$, then β is equal to

[JEE (Main)-2020]

(1) $\frac{2}{\sqrt{3}}$ (2) $\frac{2\sqrt{2}}{3}$ (3) $\frac{2}{3}$ (4) $\frac{\sqrt{2}}{3}$ **EL0094**

15. The length of the minor axis (along y-axis) of an ellipse in the standard form is $\frac{4}{\sqrt{3}}$. If this ellipse touches the line, $x + 6y = 8$; then its eccentricity is : **[JEE (Main)-2020]**

(1) $\sqrt{\frac{5}{6}}$ (2) $\frac{1}{2}\sqrt{\frac{11}{3}}$ (3) $\frac{1}{3}\sqrt{\frac{11}{3}}$ (4) $\frac{1}{2}\sqrt{\frac{5}{3}}$

EL0095

16. If the point of intersections of the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the circle $x^2 + y^2 = 4b$, $b > 4$ lie on the curve $y^2 = 3x^2$, then b is equal to: **[JEE (Main)-2021]**

(1) 12 (2) 5 (3) 6 (4) 10

EL0128

17. Let L be a tangent line to the parabola $y^2 = 4x - 20$ at (6, 2). If L is also a tangent to the ellipse $\frac{x^2}{2}+\frac{y^2}{b}=1$, then the value of b is equal to : **[JEE (Main)-2021]**

(1) 11 (2) 14 (3) 16 (4) 20

EL0129

18. If m is the slope of a common tangent to the curves $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $x^2 + y^2 = 12$, then $12m^2$ is equal to : **[JEE (Main)-2022]**

(A) 6 (B) 9 (C) 10 (D) 12

EL0130

19. The locus of the mid point of the line segment joining the point (4, 3) and the points on the ellipse $x^2 + 2y^2 = 4$ is an ellipse with eccentricity : **[JEE (Main)-2022]**

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

EL0131

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\5-Conic Section\Eng\02. Ellipse.p65

EXERCISE (JA)

1. Suppose that the foci of the ellipse $\frac{x^2}{9}+\frac{y^2}{5}=1$ are $(f_1,0)$ and $(f_2,0)$ where $f_1>0$ and $f_2<0$. Let P_1 and P_2 be two parabolas with a common vertex at (0,0) and with foci at $(f_1,0)$ and $(2f_2,0)$, respectively. Let T_1 be a tangent to P_1 which passes through $(2f_2,0)$ and T_2 be a tangent to P_2 which passes through $(f_1,0)$. If m_1 is the slope of T_1 and m_2 is the slope of T_2, then the value of $\left(\frac{1}{m_1^2}+m_2^2\right)$ is

[JEE 2015, 4M, –0M]

EL0103

2. Let E_1 and E_2 be two ellipses whose centers are at the origin. The major axes of E_1 and E_2 lie along the x-axis and the y-axis, respectively. Let S be the circle $x^2 + (y - 1)^2 = 2$. The straight line $x+y=3$ touches the curves S, E_1 and E_2 at P,Q and R, respectively. Suppose that $PQ = PR = \frac{2\sqrt{2}}{3}$. If e_1 and e_2 are the eccentricities of E_1 and E_2, respectively, then the correct expression(s) is(are)

[JEE 2015, 4M, –0M]

(A) $e_1^2+e_2^2=\frac{43}{40}$ (B) $e_1e_2=\frac{\sqrt{7}}{2\sqrt{10}}$

(C) $\left|e_1^2-e_2^2\right|=\frac{5}{8}$ (D) $e_1e_2=\frac{\sqrt{3}}{4}$

EL0104

PARAGRAPH :

Let $F_1(x_1, 0)$ and $F_2(x_2, 0)$ for $x_1 < 0$ and $x_2 > 0$, be the foci of the ellipse $\frac{x^2}{9}+\frac{y^2}{8}=1$. Suppose a parabola having vertex at the origin and focus at F_2 intersects the ellipse at point M in the first quadrant and at point N in the fourth quadrant.

3. The orthocentre of the triangle F_1MN is- **[JEE(Advanced)-2016, 4(–2)]**

(A) $\left(-\frac{9}{10},0\right)$ (B) $\left(\frac{2}{3},0\right)$ (C) $\left(\frac{9}{10},0\right)$ (D) $\left(\frac{2}{3},\sqrt{6}\right)$ **EL0105**

4. If the tangents to the ellipse at M and N meet at R and the normal to the parabola at M meets the x-axis at Q, then the ratio of area of the triangle MQR to area of the quadrilateral MF_1NF_2 is -

[JEE(Advanced)-2016, 3(0)]

(A) 3 : 4 (B) 4 : 5 (C) 5 : 8 (D) 2 : 3 **EL0105**

5. Consider two straight lines, each of which is tangent to both the circle $x^2 + y^2 = \frac{1}{2}$ and the parabola $y^2 = 4x$. Let these lines intersect at the point Q. Consider the ellipse whose center is at the origin O(0, 0) and whose semi-major axis is OQ. If the length of the minor axis of this ellipse is $\sqrt{2}$, then the which of the following statement(s) is (are) TRUE ? **[JEE(Advanced)-2018]**

(A) For the ellipse, the eccentricity is $\frac{1}{\sqrt{2}}$ and the length of the latus rectum is 1 **EL0111**

(B) For the ellipse, the eccentricity is $\frac{1}{2}$ and the length of the latus rectum is $\frac{1}{2}$

(C) The area of the region bounded by the ellipse between the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$ is $\frac{1}{4\sqrt{2}}(\pi - 2)$

(D) The area of the region bounded by the ellipse between the lines $x = \frac{1}{\sqrt{2}}$ and $x = 1$ is $\frac{1}{16}(\pi - 2)$

6. Define the collections $\{E_1, E_2, E_3, ...\}$ of ellipses and $\{R_1, R_2, R_3, ...\}$ of rectangles as follows :

$E_1 : \frac{x^2}{9} + \frac{y^2}{4} = 1$;

R_1 : rectangle of largest area, with sides parallel to the axes, inscribed in E_1 ;

E_n : ellipse $\frac{x^2}{a_n^2} + \frac{y^2}{b_n^2} = 1$ of largest area inscribed in R_{n-1}, $n > 1$;

R_n : rectangle of largest area, with sides parallel to the axes, inscribed in E_n, $n > 1$.

Then which of the following options is/are correct ? **[JEE(Advanced)-2019, 4(–1)]**

(1) The eccentricities of E_{18} and E_{19} are NOT equal

(2) The distance of a focus from the centre in E_9 is $\frac{\sqrt{5}}{32}$

(3) The length of latus rectum of E_9 is $\frac{1}{6}$

(4) $\sum_{n=1}^{N}(\text{area of } R_n) < 24$, for each positive integer N **EL0107**

7. Let a, b and λ be positive real numbers. Suppose P is an end point of the latus rectum of the parabola $y^2 = 4\lambda x$, and suppose the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the point P. If the tangents to the parabola and the ellipse at the point P are perpendicular to each other, then the eccentricity of the ellipse is **[JEE(Advanced)-2020]**

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) $\frac{1}{3}$ (D) $\frac{2}{5}$

EL0112

8. Let E be the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$. For any three distinct points P, Q and Q' on E, let M (P, Q) be the mid-point of the line segment joining P and Q, and M (P, Q') be the mid-point of the line segment joining P and Q'. Then the maximum possible value of the distance between M(P, Q) and M(P, Q'), as P, Q and Q' vary on E, is ____. **[JEE(Advanced)-2021]**

EL0113

9. Consider the ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$. Let H(α, 0), 0 < α < 2, be a point. A straight line drawn through H parallel to the y-axis crosses the ellipse and its auxiliary circle at points E and F respectively, in the first quadrant. The tangent to the ellipse at the point E intersects the positive x-axis at a point G. Suppose the straight line joining F and the origin makes an angle φ with the positive x-axis.

List-I	**List-II**
(I) If $\phi=\frac{\pi}{4}$, then the area of the triangle FGH is	(P) $\frac{\left(\sqrt{3}-1\right)^4}{8}$
(II) If $\phi=\frac{\pi}{3}$, then the area of the triangle FGH is	(Q) 1
(III) If $\phi=\frac{\pi}{6}$, then the area of the triangle FGH is	(R) $\frac{3}{4}$
(IV) If $\phi=\frac{\pi}{12}$, then the area of the triangle FGH is	(S) $\frac{1}{2\sqrt{3}}$
	(T) $\frac{3\sqrt{3}}{2}$

The correct option is : **[JEE(Advanced)-2022]**

(A) (I) → (R); (II) → (S); (III) → (Q); (IV) → (P)

(B) (I) → (R); (II) → (T); (III) → (S); (IV) → (P)

(C) (I) → (Q); (II) → (T); (III) → (S); (IV) → (P)

(D) (I) → (Q); (II) → (S); (III) → (Q); (IV) → (P)

EL0132

ANSWER KEY

Do yourself - 1

1. $e=\frac{1}{\sqrt{2}}$ **2.** $\frac{(x-10)^2}{52}+\frac{(y-6)^2}{16}=1$ **3.** $e=\frac{\sqrt{3}}{2}$; foci $=(1\pm\sqrt{3}, -1)$; LR $=1$

4. (a) $3x^2 + 5y^2 = 32$; (b) $3x^2 + 7y^2 = 115$

5. (a) $20x^2 + 36y^2 = 405$ (b) $x^2 + 2y^2 = 100$ (c) $8x^2 + 9y^2 = 1152$

Do yourself - 2

1. $\frac{x^2}{4}+\frac{y^2}{8}=1$

2. $C \equiv (-1, 2)$, length of major axis $= 2b = \sqrt{3}$, length of minor axis $= 2a = 1$; $e=\sqrt{\frac{2}{3}}$;

$f\left(-1,\ 2\pm\frac{1}{\sqrt{2}}\right)$

3. C

4. $7x^2 + 2xy + 7y^2 + 10x - 10y + 7 = 0$

Do yourself - 3

1. On the ellipse **2.** $\frac{1}{2}(a^2+b^2)$ **3.** $P^2 = a^2\cos^2\theta + b^2\sin^2\theta$

4. Without **5.** $x+4\sqrt{3}y=24\sqrt{3}; 11x-4\sqrt{3}y=24\sqrt{3};$ 7 and 13

6. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{y}{b}$

Do yourself - 4

1. $3y+x\pm\sqrt{97}=0$ **2.** $7x - 12y = 50$

3. (a) $x + 3y = 5$; (b) $\pm 25x + 6y = 137$; (c) $\pm x\sqrt{7}\pm 4y=16$

5. $y=3x\pm\frac{1}{2}\sqrt{\frac{155}{3}};\left(\pm\frac{3}{26}\sqrt{65},\mp\frac{2}{39}\sqrt{195}\right)$

Do yourself - 5

1. $4x - 3y = 7$ **2.** abe **3.** -1 **4.** $\frac{\sqrt{3}}{2}$

5. (a) $9x - 3y - 5 = 0$ (b) $\pm 6x - 25y + 20 = 0$; (c) $\pm 4x \mp y\sqrt{7} = \frac{7}{4}\sqrt{7}$

Do yourself - 6

1. $\frac{x}{16} + \frac{y}{3} = 1$ **2.** $-\frac{a^4}{b^4}$ **3.** $(12, -2)$

4. $x - 2y + 3 = 0$ **5.** 7.00

Do yourself - 7

1. $-9x + 16y = 25$

2. (a) $(b^2x^2 + a^2y^2)^2 = c^2(b^4x^2 + a^4y^2)$.

(b) $(a^2 + b^2)(b^2x^2 + a^2y^2)^2 = a^2b^2(b^4x^2 + a^4y^2)$.

(c) $b^2x(x - h) + a^2y(y - k) = 0$.

(d) $(a^4y^2 + b^4x^2)(b^2x^2 + a^2y^2 - a^2b^2) + c^2a^2b^2(a^2y^2 + b^2x^2) = 0$

3. $\frac{\pi}{2}$ **4.** $y = -\frac{x}{2} + 2$ and $x = 2$ **5.** $\frac{2\sqrt{17}}{3}$

Do yourself - 8

1. 60π **2.** $\frac{\alpha^2}{a^4} + \frac{\beta^2}{b^2} = \frac{1}{k^2}$ **3.** $\frac{2\sqrt{2}}{3}$

5. $\frac{7}{13}$ **6.** $\frac{14}{\sqrt{3}}$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	C	C	D	C	A	B	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	(a) D; (b) A,B,C,D	A	A	B	D	A	B	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	A	C	B	A	B	C	D	A	B	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,B,D	A,C	A,C	A,B	A,C,D	A,C,D	C,D	A,D	B,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B,D	A,B	A,C	A,B	A,B	A,C	A,B,C	A,D	A,B,C	B,C

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	C	B	D	D	C	A	46.00	8.50	A
Q.10	A	B	C	D					
	S	R	P	Q					

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	60	80	1.05 or 1.06	2.00	64.00	3.00	8.00	5.00	0.50

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	1	4	3	2	3	3	3	1
Que.	11	12	13	14	15	16	17	18	19	
Ans.	4	2	3	4	2	1	2	B	C	

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9
Ans.	4	A,B	A	C	A,C	3,4	A	4	C

EXERCISE (O-1)

Straight Objective Type

1. Eccentricity of the hyperbola conjugate to the hyperbola $\frac{x^2}{4}-\frac{y^2}{12}=1$ is

(A) $\frac{2}{\sqrt{3}}$ (B) 2 (C) $\sqrt{3}$ (D) $\frac{4}{3}$ **HB0003**

2. The foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$. Then the equation of the hyperbola with eccentricity 2 is

(A) $\frac{x^2}{12}-\frac{y^2}{4}=1$ (B) $\frac{x^2}{4}-\frac{y^2}{12}=1$

(C) $3x^2 - y^2 + 12 = 0$ (D) $9x^2 - 25y^2 - 225 = 0$ **HB0004**

3. If the eccentricity of the hyperbola $x^2 - y^2 \sec^2 \alpha = 5$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2 \sec^2 \alpha + y^2 = 25$, then a value of α is :

(A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $\pi/2$ **HB0005**

4. The foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ coincide. Then the value of b^2 is-

(A) 5 (B) 7 (C) 9 (D) 4 **HB0006**

5. The graph of the equation $x + y = x^3 + y^3$ is the union of -

(A) line and an ellipse (B) line and a parabola

(C) line and hyperbola (D) line and a point **HB0007**

6. The focal length of the hyperbola $x^2 - 3y^2 - 4x - 6y - 11 = 0$, is -

(A) 4 (B) 6 (C) 8 (D) 10 **HB0008**

7. The equation $\frac{x^2}{29-p}+\frac{y^2}{4-p}=1$ $(p \neq 4, 29)$ represents -

(A) an ellipse if p is any constant greater than 4

(B) a hyperbola if p is any constant between 4 and 29.

(C) a rectangular hyperbola if p is any constant greater than 29.

(D) no real curve is p is less than 29. **HB0009**

8. If $\frac{x^2}{\cos^2 \alpha}-\frac{y^2}{\sin^2 \alpha}=1$ represents family of hyperbolas where 'α' varies then -

(A) distance between the foci is constant

(B) distance between the two directrices is constant

(C) distance between the vertices is constant

(D) distances between focus and the corresponding directrix is constant **HB0010**

9. The locus of the point of intersection of the lines $\sqrt{3}x-y-4\sqrt{3}\,t=0$ & $\sqrt{3}tx+ty-4\sqrt{3}=0$ (where t is a parameter) is a hyperbola whose eccentricity is

(A) $\sqrt{3}$ (B) 2 (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{4}{3}$ **HB0011**

10. Latus rectum of the conic satisfying the differential equation, $x\,dy+y\,dx=0$ and passing through the point (2, 8) is :

(A) $4\sqrt{2}$ (B) 8 (C) $8\sqrt{2}$ (D) 16 **HB0012**

11. The magnitude of the gradient of the tangent at an extremity of latera recta of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is equal to (where e is the eccentricity of the hyperbola)

(A) be (B) e (C) ab (D) ae **HB0013**

12. The number of possible tangents which can be drawn to the curve $4x^2-9y^2=36$, which are perpendicular to the straight line $5x+2y-10=0$ is :

(A) zero (B) 1 (C) 2 (D) 4 **HB0014**

13. Locus of the point of intersection of the tangents at the points with eccentric angles ϕ and $\frac{\pi}{2}-\phi$ on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is :

(A) $x=a$ (B) $y=b$ (C) $x=ab$ (D) $y=ab$ **HB0015**

14. Locus of the feet of the perpendiculars drawn from either foci on a variable tangent to the hyperbola $16y^2-9x^2=1$ is

(A) $x^2+y^2=9$ (B) $x^2+y^2=1/9$

(C) $x^2+y^2=7/144$ (D) $x^2+y^2=1/16$ **HB0016**

15. A tangent to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ with centre C meets its director circle at P and Q. Then the product of the slopes of CP and CQ, is -

(A) $\frac{9}{4}$ (B) $\frac{-4}{9}$ (C) $\frac{2}{9}$ (D) $-\frac{1}{4}$ **HB0017**

16. The asymptote of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ form with any tangent to the hyperbola a triangle whose area is $a^2\tan\lambda$ in magnitude then its eccentricity is :

(A) $\sec\lambda$ (B) $\text{cosec}\lambda$ (C) $\sec^2\lambda$ (D) $\text{cosec}^2\lambda$ **HB0018**

17. In which of the following cases maximum number of normals can be drawn from a point P lying in the same plane

(A) circle (B) parabola (C) ellipse (D) hyperbola

HB0019

18. PQ is a double ordinate of the ellipse $x^2 + 9y^2 = 9$, the normal at P meets the diameter through Q at R, then the locus of the mid point of PR is

(A) a circle (B) a parabola (C) an ellipse (D) a hyperbola

HB0020

19. With one focus of the hyperbola $\frac{x^2}{9} - \frac{y^2}{16} = 1$ as the centre , a circle is drawn which is tangent to the hyperbola with no part of the circle being outside the hyperbola. The radius of the circle is

(A) less than 2 (B) 2 (C) $\frac{11}{3}$ (D) none **HB0021**

20. If the normal to the rectangular hyperbola $xy = c^2$ at the point 't' meets the curve again at 't_1' then $t^3\, t_1$ has the value equal to

(A) 1 (B) – 1 (C) 0 (D) none **HB0022**

21. Number of common tangent with finite slope to the curves $xy = c^2$ & $y^2 = 4ax$ is :

(A) 0 (B) 1 (C) 2 (D) 4 **HB0023**

22. Locus of the middle points of the parallel chords with gradient m of the rectangular hyperbola $xy = c^2$ is

(A) $y + mx = 0$ (B) $y - mx = 0$ (C) $my - x = 0$ (D) $my + x = 0$

HB0024

23. Let F_1, F_2 are the foci of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ and F_3, F_4 are the foci of its conjugate hyperbola. If e_H and e_C are their eccentricities respectively then the statement which holds true is

(A) Their equations of the asymptotes are different.

(B) $e_H > e_C$

(C) Area of the quadrilateral formed by their foci is 50 sq. units.

(D) Their auxiliary circles will have the same equation. **HB0041**

24. AB is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that ΔAOB (where 'O' is the origin) is an equilateral triangle, then the eccentricity e of the hyperbola satisfies

(A) $e > \sqrt{3}$ (B) $1 < e < \frac{2}{\sqrt{3}}$ (C) $e = \frac{2}{\sqrt{3}}$ (D) $e > \frac{2}{\sqrt{3}}$ **HB0043**

25. P is a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, N is the foot of the perpendicular from P on the transverse axis. The tangent to the hyperbola at P meets the transverse axis at T . If O is the centre of the hyperbola, then OT.ON is equal to :

(A) e^2 (B) a^2 (C) b^2 (D) b^2/a^2 **HB0044**

26. Let the major axis of a standard ellipse equals the transverse axis of a standard hyperbola and their director circles have radius equal to 2R and R respectively. If e_1 and e_2 are the eccentricities of the ellipse and hyperbola then the correct relation is

(A) $4e_1^2 - e_2^2 = 6$ (B) $e_1^2 - 4e_2^2 = 2$ (C) $4e_2^2 - e_1^2 = 6$ (D) $2e_1^2 - e_2^2 = 4$

HB0045

27. The tangent to the hyperbola $xy = c^2$ at the point P intersects the x-axis at T and the y-axis at T′. The normal to the hyperbola at P intersects the x-axis at N and the y-axis at N′. The areas of the triangles PNT and PN'T' are Δ and Δ' respectively, then $\frac{1}{\Delta} + \frac{1}{\Delta'}$ is

(A) equal to 1 (B) depends on t (C) depends on c (D) equal to 2

HB0046

28. The chord PQ of the rectangular hyperbola $xy = a^2$ meets the axis of x at A ; C is the mid point of PQ & 'O' is the origin. Then the Δ ACO is :

(A) equilateral (B) isosceles (C) right angled (D) right isosceles.

HB0047

29. The locus of the foot of the perpendicular from the centre of the hyperbola $xy = c^2$ on a variable tangent is :

(A) $(x^2 - y^2)^2 = 4c^2 xy$ (B) $(x^2 + y^2)^2 = 2c^2 xy$

(C) $(x^2 + y^2) = 4c^2 xy$ (D) $(x^2 + y^2)^2 = 4c^2 xy$ **HB0048**

30. The equation to the chord joining two points (x_1, y_1) and (x_2, y_2) on the rectangular hyperbola $xy = c^2$ is :

(A) $\frac{x}{x_1 + x_2} + \frac{y}{y_1 + y_2} = 1$ (B) $\frac{x}{x_1 - x_2} + \frac{y}{y_1 - y_2} = 1$

(C) $\frac{x}{y_1 + y_2} + \frac{y}{x_1 + x_2} = 1$ (D) $\frac{x}{y_1 - y_2} + \frac{y}{x_1 - x_2} = 1$ **HB0049**

EXERCISE (O-2)

Multiple Correct Answer Type

1. Let p and q be non-zero real numbers. Then the equation $(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$ represents

(A) two straight lines and a circle, when r = 0 and p, q are of the opposite sign.

(B) two circles, when p = q and r is of sign opposite to that of p.

(C) a hyperbola and a circle, when p and q are of opposite sign and $r \neq 0$.

(D) a circle and an ellipse, when p and q are unequal but of same sign and r is of sign opposite to that of p. **HB0025**

2. If θ is eliminated from the equations $a \sec\theta - x\tan\theta = y$ and $b\sec\theta + y\tan\theta = x$ (a and b are constant), then the eliminant denotes the equation of

(A) the director circle of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(B) auxiliary circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(C) Director circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(D) Director circle of the circle $x^2 + y^2 = \frac{a^2 + b^2}{2}$. **HB0027**

3. The tangent to the hyperbola, $x^2 - 3y^2 = 3$ at the point $\left(\sqrt{3}, 0\right)$ when associated with two asymptotes constitutes -

(A) isosceles triangle which is not equilateral (B) an equilateral triangle

(C) a triangles whose area is $\sqrt{3}$ sq. units (D) a right isosceles triangle. **HB0028**

4. If latus rectum of a hyperbola subtends a right angle at other focus of hyperbola, then eccentricity is equal to-

(A) $1-\sqrt{2}$ (B) $\tan\frac{\pi}{8}$ (C) $\cot\frac{\pi}{8}$ (D) $\left(\frac{1}{\sqrt{2}-1}\right)$

HB0029

5. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points $P(x_1,y_1)$, $Q(x_2,y_2)$, $R(x_3,y_3)$, $S(x_4,y_4)$, then -

(A) $x_1 + x_2 + x_3 + x_4 = 0$ (B) $y_1 + y_2 + y_3 + y_4 = 0$

(C) $x_1x_2x_3x_4 = c^4$ (D) $y_1y_2y_3y_4 = c^4$ **HB0030**

6. Which of the following equations in parametric form can represent a hyperbolic profile, where 't' is a parameter.

(A) $x=\frac{a}{2}\left(t+\frac{1}{t}\right)$ & $y=\frac{b}{2}\left(t-\frac{1}{t}\right)$

(B) $\frac{tx}{a}-\frac{y}{b}+t=0$ & $\frac{x}{a}+\frac{ty}{b}-1=0$

(C) $x=e^t+e^{-t}$ & $y=e^t-e^{-t}$

(D) $x^2-6=2\cos t$ & $y^2+2=4\cos^2\frac{t}{2}$

HB0051

7. Let A(–1,0) and B(2,0) be two points on the x-axis. A point M is moving in xy-plane (other than x-axis) in such a way that ∠MBA = 2∠MAB, then the point M moves along a conic whose

(A) coordinate of vertices are (± 3, 0).

(B) length of latus-rectum equals 6.

(C) eccentricity equals 2.

(D) equation of directrices are $x=\pm\frac{1}{2}$.

HB0052

8. Hyperbola $\frac{x^2}{a^2}-\frac{y^2}{3}=1$ of eccentricity e is confocal with the ellipse $\frac{x^2}{8}+\frac{y^2}{4}=1$. Let A,B,C & D are points of intersection of hyperbola & ellipse, then-

(A) $e=\frac{5}{2}$

(B) e = 2

(C) A,B,C,D are concyclic points

(D) Number of common tangents of hyperbola & ellipse is 2

HB0053

9. If the ellipse $4x^2+9y^2+12x+12y+5=0$ is confocal with a hyperbola having same principal axes, then -

(A) angle between normals at their each point of intersection is 90º.

(B) centre of the ellipse is $\left(-\frac{3}{2},-\frac{2}{3}\right)$

(C) distance between foci of the hyperbola is $\frac{2\sqrt{10}}{3}$

(D) ellipse and hyperbola has same length of latus rectum

HB0054

10. If the eccentricity of the ellipse $\frac{x^2}{(\log a)^2}+\frac{y^2}{(\log b)^2}=1$ (a > b > 0, a, b ≠ 1) is $\frac{1}{\sqrt{2}}$ and 'e' be the eccentricity of the hyperbola $\frac{x^2}{(\log_b a)^2}-y^2=1$, then e^2 is greater than (where logx = ℓnx)-

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{5}{4}$

HB0055

11. Let the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be reciprocal to that of the ellipse $x^2+4y^2=4$. If the hyperbola passes through a focus of the ellipse, then -

(A) the equation of the hyperbola is $\frac{x^2}{3}-\frac{y^2}{2}=1$ (B) a focus of the hyperbola is (2, 0)

(C) the eccentricity of the hyperbola is $\sqrt{\frac{5}{3}}$ (D) the equation of the hyperbola is $x^2-3y^2=3$

HB0110

12. Tangents are drawn to the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$, parallel to the straight line $2x-y=1$. The points of contact of the tangents on the hyperbola are

(A) $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (B) $\left(-\frac{9}{2\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ (C) $\left(3\sqrt{3}, -2\sqrt{2}\right)$ (D) $\left(-3\sqrt{3}, 2\sqrt{2}\right)$

HB0112

13. An ellipse intersects the hyperbola $2x^2-2y^2=1$ orthogonally. The eccentricity of the ellipse is reciprocal of that of the hyperbola. If the axes of the ellipse are along the coordinate axes, then

(A) Equation of ellipse is $x^2+2y^2=2$ (B) The foci of ellipse are $(\pm 1, 0)$

(C) Equation of ellipse is $x^2+2y^2=4$ (D) The foci of ellipse are $(\pm\sqrt{2}, 0)$

HB0102

14. If the normal at point P to the rectangular hyperbola $x^2-y^2=4$ meets the transverse and conjugate axes at A and B respectively and C is the centre of the hyperbola, then -

(A) PA = PC (B) PA = PB (C) PB = PC (D) AB = 2PC

HB0119

15. Let an incident ray $L_1=0$ gets reflected at point A(–2, 3) on hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and passes through focus S(2, 0), then -

(A) equation of incident ray is $x+2=0$ (B) equation of reflected ray is $3x+4y=6$

(C) eccentricity, e = 2 (D) length of latus rectum = 6

HB0120

16. All the points on the asymptotes of a certain hyperbola satisfy the equation of $x^2-4y^2=0$ then the possible value(s) of eccentricity of the hyperbola is/are -

(A) $\frac{\sqrt{5}}{2}$ (B) $\frac{\sqrt{5}}{3}$ (C) $\sqrt{5}$ (D) $\frac{3}{2}$ **HB0121**

17. If the equation of a hyperbola is given as $\left|\sqrt{(x-3)^2+(y-3)^2}-\sqrt{(x-1)^2+(y-1)^2}\right|=2$, then-

(A) The given hyperbola is a rectangular hyperbola.

(B) Distance between both directrices is $\sqrt{2}$

(C) Centre of hyperbola lies on line $y = x$

(D) Asymptotes are inclined at an angle of 45° with each other. **HB0122**

18. Let the variable line $y = kx + h$ is tangent to the hyperbola $\frac{x^2}{4}-\frac{y^2}{9}=1$. If locus of P(h,k) is a conic then which of the following is true ?

(A) focus of conic lies on y-axis.

(B) eccentricity of conic is greater than the eccentricity of given hyperbola

(C) centre of conic is at origin

(D) conic is an ellipse **HB0123**

19. If the ellipse $x^2 + 2y^2 = 4$ touches the hyperbola $xy = c$ at points P and Q, then -

(A) c is a rational number

(B) O,P and Q are collinear points (where 'O' is origin)

(C) length of PQ is $2\sqrt{3}$

(D) length of PQ is $2\sqrt{2}$

HB0124

20. If (3,4) and (8,6) are foci of a conic passing through origin, then eccentricity of conic is-

(A) $\frac{\sqrt{29}}{16}$ (B) $\frac{\sqrt{29}}{15}$ (C) $\frac{\sqrt{29}}{12}$ (D) $\frac{\sqrt{29}}{5}$

HB0125

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for question nos. 1 to 3

From a point 'P' three normals are drawn to the parabola $y^2 = 4x$ such that two of them make angles with the abscissa axis, the product of whose tangents is 2. Suppose the locus of the point 'P' is a part of a conic 'C'. Now a circle S = 0 is described on the chord of the conic 'C' as diameter passing through the point (1, 0) and with gradient unity. Suppose (a, b) are the coordinates of the centre of this circle. If L_1 and L_2 are the two asymptotes of the hyperbola with length of its transverse axis 2a and conjugate axis 2b (principal axes of the hyperbola along the coordinate axes) then answer the following questions.

1. Locus of P is a

(A) circle (B) parabola (C) ellipse (D) hyperbola

HB0056

2. Radius of the circle S = 0 is

(A) 4 (B) 5 (C) $\sqrt{17}$ (D) $\sqrt{23}$ **HB0056**

3. The angle $\alpha \in (0, \pi/2)$ between the two asymptotes of the hyperbola lies in the interval

(A) (0, 15°) (B) (30°, 45°) (C) (45°, 60°) (D) (60°, 75°)

HB0056

Paragraph for question nos. 4 to 6

Consider the conic C : $\frac{x^2}{16}+\frac{y^2}{12}=1$

4. Equation of circle touching C at one extremity of latus-rectum and passing through centre of C is/are-

(A) $8x^2 + 8y^2 - 19x - 22y = 0$ (B) $8x^2 + 8y^2 - 19x + 22y = 0$

(C) $8x^2 + 8y^2 + 19x - 22y = 0$ (D) $8x^2 + 8y^2 + 19x + 22y = 0$ **HB0032**

5. The equation of parabolas with same latus-rectum as conic C, is/are-

(A) $y^2 - 6x + 3 = 0$ (B) $y^2 + 6x - 21 = 0$ (C) $y^2 - 6x - 21 = 0$ (D) $y^2 + 6x + 3 = 0$

HB0032

6. If a hyperbola passes through extremities of minor axis of conic C and its transverse and conjugate axis coincides with the minor and major axis of conic C respectively, and the product of eccentricity of hyperbola and conic C is 1, then -

(A) equation of hyperbola is $\frac{x^2}{36}-\frac{y^2}{12}=-1.$ (B) equation of hyperbola is $\frac{x^2}{9}-\frac{y^2}{3}=-1.$

(C) focus of hyperbola is $(0, 2\sqrt{3})$ (D) focus of hyperbola is $(0, -4\sqrt{3})$ **HB0032**

Paragraph for question nos. 7 to 8

The graph of the conic $x^2 - (y - 1)^2 = 1$ has one tangent line with positive slope that passes through the origin. the point of tangency being (a, b). Then

7. The value of $\sin^{-1}\left(\frac{a}{b}\right)$ is $\lambda\pi$, then λ is **HB0031**

8. Length of the latus rectum of the conic is **HB0031**

9. **List-I** **List-II**

(P) Value of c for which $3x^2 - 5xy - 2y^2 + 5x + 11y + c = 0$ are the asymptotes of the hyperbola $3x^2 - 5xy - 2y^2 + 5x + 11y - 8 = 0$ — (1) 3

(Q) If locus of a point, whose chord of contact with respect to the circle $x^2 + y^2 = 4$ is a tangent to the hyperbola $xy = 1$ is $xy = c^2$, then value of c^2 is — (2) –4

(R) If equation of a hyperbola whose conjugate axis is 5 and distance between its foci is 13, is $ax^2 - by^2 = c$ where a and b are coprime natural numbers, then value of $\frac{ab}{c}$ is — (3) –12

(S) If the vertex of a hyperbola bisects the distance between its centre and the corresponding focus, then ratio of square of its conjugate axis to the square of its transverse axis is — (4) 4

(5) –6

Which of the following is correct option ?

(A) $P \to 2; Q \to 1; R \to 4; S \to 1$ (B) $P \to 3; Q \to 4; R \to 1; S \to 5$

(C) $P \to 3; Q \to 4; R \to 4; S \to 1$ (D) $P \to 1; Q \to 4; R \to 5; S \to 3$

HB0126

10. **Column-I** **Column-II**

(A) Number of positive integral values of b for which tangent parallel to line $y = x + 1$ can be drawn to hyperbola $\frac{x^2}{5} - \frac{y^2}{b^2} = 1$ is — (P) 16

(B) The equation of the hyperbola with vertices (3, 0) and (–3, 0) and semi-latusrectum 4, is given by is $4x^2 - 3y^2 = 4k$, then $k =$ — (Q) 2

(C) The product of the lengths of the perpendiculars from the two focii on any tangent to the hyperbola $\frac{x^2}{25} - \frac{y^2}{3} = 1$ is $\sqrt{k}$, then k is — (R) 4

(D) An equation of a tangent to the hyperbola, $16x^2 - 25y^2 - 96x + 100y - 356 = 0$ which makes an angle $\frac{\pi}{4}$ with the transverse axis is $y = x + \lambda, (\lambda > 0)$, then 2λ is — (S) 9

HB0127

EXERCISE (O-4)

Numerical Grid Type

1. The hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ (a, b > 0) passes through the point of intersection of the lines, $7x + 13y - 87 = 0$ & $5x - 8y + 7 = 0$ and the latus rectum is $\frac{32\sqrt{2}}{5}$, then value of b is equal to **HB0128**

2. The line 2x + y = 1 is tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If this line passes through the point of intersection of the nearest directrix and the x-axis, then the eccentricity of the hyperbola is **HB0109**

3. Tangents to the hyperbola $x^2 - 9y^2 = 9$ that are drawn from (3, 2). Find the area of the triangle that these tangents form with their chord of contact. **HB0077**

4. Let 'p' be the perpendicular distance from the centre C of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ to the tangent drawn at a point R on the hyperbola. If S & S′ are the two foci of the hyperbola and $(RS + RS')^2 = \lambda a^2\left(1+\frac{b^2}{p^2}\right)$, then find λ. **HB0072**

5. From the centre C of the hyperbola $x^2 - y^2 = 9$, CM is drawn perpendicular to the tangent at any point of the curve, meeting the tangent at M and the curve at N. Find the value of the product (CM)(CN). **HB0080**

6. Find the equation to the hyperbola whose directrix is 2x + y = 1, focus (1, 1) & eccentricity $\sqrt{3}$. If length of its latus rectum is $\sqrt{\frac{p}{q}}$ then (p + q) is **HB0060**

7. The hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ passes through the point of intersection of the lines, $7x + 13y - 87 = 0$ and $5x - 8y + 7 = 0$ & the latus rectum is $32\sqrt{2}/5$. Find $(2a^2 + b^2)$. **HB0061**

8. If the lines x + y + 1 = 0 and 2x – y + 2 = 0 are the asymptotes of a hyperbola. If the line x – 2 = 0 touches the hyperbola then the equation of the hyperbola is $4(x + y + 1)(2x - y + 2) = \lambda$. Find the value of λ. **HB0069**

9. Locus of the feet of the perpendicular from centre of the hyperbola $x^2 - 4y^2 = 4$ upon a variable normal to it has the equation, $(x^2 + y^2)^2 (4y^2 - x^2) = \lambda x^2y^2$, find λ. **HB0074**

10. Tangent and normal are drawn at the upper end (x_1, y_1) of the latus rectum P with $x_1 > 0$ and $y_1 > 0$, of the hyperbola $\frac{x^2}{4}-\frac{y^2}{12}=1$, intersecting the transverse axis at T and G respectively. Find the area of the triangle PTG. **HB0076**

EXERCISE (JM)

1. A tangent to the hyperbola $\frac{x^2}{4}-\frac{y^2}{2}=1$ meets x-axis at P and y-axis at Q. Lines PR and QR are drawn such that OPRQ is a rectangle (where O is the origin). Then R lies on : **[JEE-Main (On line)-2013]**

(1) $\frac{2}{x^2}-\frac{4}{y^2}=1$ (2) $\frac{4}{x^2}-\frac{2}{y^2}=1$ (3) $\frac{4}{x^2}+\frac{2}{y^2}=1$ (4) $\frac{2}{x^2}+\frac{4}{y^2}=1$

HB0087

2. A common tangent to the conics $x^2 = 6y$ and $2x^2 - 4y^2 = 9$ is : **[JEE-Main (On line)-2013]**

(1) $x + y = \frac{9}{2}$ (2) $x + y = 1$ (3) $x - y = \frac{3}{2}$ (4) $x - y = 1$

HB0088

3. The eccentricity of the hyperbola whose length of the latus rectum is equal to 8 and the length of its conjugate axis is equal to half of the distance between its foci, is : **[JEE (Main) 2016]**

(1) $\sqrt{3}$ (2) $\frac{4}{3}$ (3) $\frac{4}{\sqrt{3}}$ (4) $\frac{2}{\sqrt{3}}$ **HB0089**

4. A hyperbola passes through the point $P(\sqrt{2},\sqrt{3})$ and has foci at (± 2, 0). Then the tangent to this hyperbola at P also passes through the point : **[JEE (Main) 2017]**

(1) $(-\sqrt{2},-\sqrt{3})$ (2) $(3\sqrt{2},2\sqrt{3})$ (3) $(2\sqrt{2},3\sqrt{3})$ (4) $(\sqrt{3},\sqrt{2})$

HB0090

5. Tangents are drawn to the hyperbola $4x^2 - y^2 = 36$ at the point P and Q. If these tangents intersect at the point T(0, 3) then the area (in sq. units) of ΔPTQ is - **[JEE (Main) 2018]**

(1) $54\sqrt{3}$ (2) $60\sqrt{3}$ (3) $36\sqrt{5}$ (4) $45\sqrt{5}$ **HB0091**

6. Let $S=\left\{(x,y)\in R^2:\frac{y^2}{1+r}-\frac{x^2}{1-r}=1\right\}$, where $r \neq \pm 1$. Then S represents : **[JEE (Main)-2019]**

(1) A hyperbola whose eccentricity is $\frac{2}{\sqrt{r+1}}$, where $0 < r < 1$.

(2) An ellipse whose eccentricity is $\frac{1}{\sqrt{r+1}}$, where $r > 1$

(3) A hyperbola whose eccentricity is $\frac{2}{\sqrt{1-r}}$, when $0 < r < 1$.

(4) An ellipse whose eccentricity is $\sqrt{\frac{2}{r+1}}$, when $r > 1$ **HB0092**

7. Equation of a common tangent to the parabola $y^2 = 4x$ and the hyperbole $xy = 2$ is : **[JEE (Main)-2019]**

(1) $x + 2y + 4 = 0$ (2) $x - 2y + 4 = 0$

(3) $x + y + 1 = 0$ (4) $4x + 2y + 1 = 0$ **HB0093**

8. If a hyperbola has length of its conjugate axis equal to 5 and the distance between its foci is 13, then the eccentricity of the hyperbola is :- **[JEE (Main)-2019]**

(1) 2 (2) $\frac{13}{6}$ (3) $\frac{13}{8}$ (4) $\frac{13}{12}$ **HB0094**

9. If the line $y = mx + 7\sqrt{3}$ is normal to the hyperbola $\frac{x^2}{24} - \frac{y^2}{18} = 1$, then a value of m is **[JEE (Main)-2019]**

(1) $\frac{\sqrt{5}}{2}$ (2) $\frac{3}{\sqrt{5}}$ (3) $\frac{2}{\sqrt{5}}$ (4) $\frac{\sqrt{15}}{2}$ **HB0095**

10. If a directrix of a hyperbola centred at the origin and passing through the point $\left(4, -2\sqrt{3}\right)$ is $5x = 4\sqrt{5}$ and its eccentricity is e, then : **[JEE (Main)-2019]**

(1) $4e^4 - 24e^2 + 35 = 0$ (2) $4e^4 + 8e^2 - 35 = 0$

(3) $4e^4 - 12e^2 - 27 = 0$ (4) $4e^4 - 24e^2 + 27 = 0$ **HB0096**

11. The equation of a common tangent to the curves, $y^2 = 16x$ and $xy = -4$ is : **[JEE (Main)-2019]**

(1) $x + y + 4 = 0$ (2) $x - 2y + 16 = 0$ (3) $2x - y + 2 = 0$ (4) $x - y + 4 = 0$

HB0117

12. Let P be the point of intersection of the common tangents to the parabola $y^2 = 12x$ and the hyperbola $8x^2 - y^2 = 8$. If S and S' denote the foci of the hyperbola where S lies on the positive x-axis then P divides SS' in a ratio : **[JEE (Main)-2019]**

(1) 5:4 (2) 14:13 (3) 2:1 (4) 13:11 **HB0097**

13. If a hyperbola passes through the point P(10,16) and it has vertices at (±6,0), then the equation of the normal to it at P is **[JEE (Main)-2020]**

(1) $x + 2y = 42$ (2) $3x + 4y = 94$ (3) $2x + 5y = 100$ (4) $x + 3y = 58$

HB0098

14. If e_1 and e_2 are the eccentricities of the ellipse, $\frac{x^2}{18} + \frac{y^2}{4} = 1$ and the hyperbola, $\frac{x^2}{9} - \frac{y^2}{4} = 1$ respectively and (e_1, e_2) is a point on the ellipse, $15x^2 + 3y^2 = k$, then k is equal to : **[JEE (Main)-2020]**

(1) 15 (2) 14 (3) 17 (4) 16 **HB0099**

15. If the curves, $\frac{x^2}{a}+\frac{y^2}{b}=1$ and $\frac{x^2}{c}+\frac{y^2}{d}=1$ intersect each other at an angle of 90°, then which of the following relations is TRUE ? **[JEE (Main)-2021]**

(1) $a+b=c+d$ (2) $a-b=c-d$ (3) $a-c=b+d$ (4) $ab=\frac{c+d}{a+b}$

HB0129

16. The locus of the point of intersection of the lines $(\sqrt{3})kx+ky-4\sqrt{3}=0$ and $\sqrt{3}x-y-4(\sqrt{3})k=0$ is a conic, whose eccentricity is ___________. **[JEE (Main)-2021]**

HB0130

17. The normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{9}=1$ at the point $(8, 3\sqrt{3})$ on it passes through the point : **[JEE (Main)-2022]**

(A) $(15, -2\sqrt{3})$ (B) $(9, 2\sqrt{3})$ (C) $(-1, 9\sqrt{3})$ (D) $(-1, 6\sqrt{3})$

HB0131

18. Let the tangent drawn to the parabola $y^2=24x$ at the point (α, β) is perpendicular to the line $2x+2y=5$. Then the normal to the hyperbola $\frac{x^2}{\alpha^2}-\frac{y^2}{\beta^2}=1$ at the point $(\alpha+4, \beta+4)$ does NOT pass through the point : **[JEE (Main)-2022]**

(A) (25, 10) (B) (20, 12) (C) (30, 8) (D) (15, 13)

HB0132

EXERCISE (JA)

1. If $2x - y + 1 = 0$ is tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{16} = 1$, then which of the following CANNOT be sides of a right angled triangle ? **[JEE(Advanced)-2017, 4(–2)]**

(A) 2a, 4, 1 (B) 2a, 8, 1 (C) a, 4, 1 (D) a, 4, 2 **HB0114**

Column 1, 2 and 3 contain conics, equation of tangents to the conics and points of contact, respectively.

Column 1	**Column 2**	**Column 3**
(I) $x^2 + y^2 = a^2$	(i) $my = m^2x + a$	(P) $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$
(II) $x^2 + a^2y^2 = a^2$	(ii) $y = mx + a\sqrt{m^2+1}$	(Q) $\left(\frac{-ma}{\sqrt{m^2+1}}, \frac{a}{\sqrt{m^2+1}}\right)$
(III) $y^2 = 4ax$	(iii) $y = mx + \sqrt{a^2m^2-1}$	(R) $\left(\frac{-a^2m}{\sqrt{a^2m^2+1}}, \frac{1}{\sqrt{a^2m^2+1}}\right)$
(IV) $x^2 - a^2y^2 = a^2$	(iv) $y = mx + \sqrt{a^2m^2+1}$	(S) $\left(\frac{-a^2m}{\sqrt{a^2m^2-1}}, \frac{-1}{\sqrt{a^2m^2-1}}\right)$

2. The tangent to a suitable conic (Column 1) at $\left(\sqrt{3}, \frac{1}{2}\right)$ is found to be $\sqrt{3}x + 2y = 4$, then which of the following options is the only **CORRECT** combination ?

[JEE(Advanced)-2017, 3(–1)]

(A) (II) (iii) (R) (B) (IV) (iv) (S) (C) (IV) (iii) (S) (D) (II) (iv) (R) **HB0115**

3. If a tangent to a suitable conic (Column 1) is found to be $y = x + 8$ and its point of contact is (8,16), then which of the following options is the only **CORRECT** combination ?

[JEE(Advanced)-2017, 3(–1)]

(A) (III) (i) (P) (B) (III) (ii) (Q) (C) (II) (iv) (R) (D) (I) (ii) (Q) **HB0115**

4. For $a = \sqrt{2}$, if a tangent is drawn to a suitable conic (Column 1) at the point of contact (–1,1), then which of the following options is the only **CORRECT** combination for obtaining its equation ?

[JEE(Advanced)-2017, 3(–1)]

(A) (II) (ii) (Q) (B) (III) (i) (P) (C) (I) (i) (P) (D) (I) (ii) (Q) **HB0115**

5. Let $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $a > b > 0$, be a hyperbola in the xy-plane whose conjugate axis LM subtends an angle of 60° at one of its vertices N. Let the area of the triangle LMN be $4\sqrt{3}$.

LIST-I	LIST-II
P. The length of the conjugate axis of H is	**1.** 8
Q. The eccentricity of H is	**2.** $\frac{4}{\sqrt{3}}$
R. The distance between the foci of H is	**3.** $\frac{2}{\sqrt{3}}$
S. The length of the latus rectum of H is	**4.** 4

The correct option is :

(A) P $\to$ 4; Q $\to$ 2, R $\to$ 1; S $\to$ 3 (B) P $\to$ 4; Q $\to$ 3; R $\to$ 1; S $\to$ 2

(C) P $\to$ 4; Q $\to$ 1, R $\to$ 3; S $\to$ 2 (D) P $\to$ 3; Q $\to$ 4; R $\to$ 2; S $\to$ 1

[JEE(Advanced)-2018, 3(–1)]

HB0116

6. Let a and b be positive real numbers such that $a > 1$ and $b < a$. Let P be a point in the first quadrant that lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Suppose the tangent to the hyperbola at P passes through the point (1,0), and suppose the normal to the hyperbola at P cuts off equal intercepts on the coordinate axes. Let Δ denote the area of the triangle formed by the tangent at P, the normal at P and the x-axis. If e denotes the eccentricity of the hyperbola, then which of the following statements is/are TRUE ?

[JEE(Advanced)-2020]

(A) $1 < e < \sqrt{2}$ (B) $\sqrt{2} < e < 2$ (C) $\Delta = a^4$ (D) $\Delta = b^4$

HB0118

7. Consider the hyperbola $\frac{x^2}{100} - \frac{y^2}{64} = 1$ with foci at S and S_1, where S lies on the positive x-axis. Let P be a point on the hyperbola, in the first quadrant. Let $\angle SPS_1 = \alpha$, with $\alpha < \frac{\pi}{2}$. The straight line passing through the point S and having the same slope as that of the tangent at P to the hyperbola, intersects the straight line S_1P at P_1. Let δ be the distance of P from the straight line SP_1, and $\beta = S_1P$. Then the greatest integer less than or equal to $\frac{\beta\delta}{9}\sin\frac{\alpha}{2}$ is______. **[JEE(Advanced)-2022]**

HB0133

ANSWER KEY

Do yourself - 1

1. $\sqrt{3}$ **2.** $7y^2 + 24xy - 24ax - 6ay + 15a^2 = 0$

3. $6, 4;\ (\pm\sqrt{13}, 0);\ \sqrt{13}/3;\ 8/3$ **4.** $x^2 - y^2 = 32$

5. **(a)** $(-4,-1)$; **(b)** $\frac{5}{4}$; **(c)** $(1,-1), (-9,-1)$; **(d)** $5x + 4 = 0,\ 5x + 36 = 0$;

(e) $\frac{9}{2}$

6. (a) $16x^2 - 9y^2 = 36$ (b) $25x^2 - 144y^2 = 900$ (c) $65x^2 - 36y^2 = 441$

7. $3x^2 - y^2 = 3a^2$ **8.** $7y^2 + 24xy - 24ax - 6ay + 15a^2 = 0;\ \left(-\frac{a}{3}, a\right); 12x - 9y + 29a = 0$

9. $\sqrt{5}$ & 40 sq. units

Do yourself - 2

1. $n^2 = a^2\ell^2 - b^2m^2$ **2.** B **3.** $(5, -\frac{20}{3})$

5. $\lambda = \pm 6$ **6.** $n = \pm 2$

Do yourself - 3

1. $24y = 30x \pm \sqrt{161}$ **2.** $5x - 3y = 9$ **3.** $y = \pm x \pm \sqrt{7}$

4. **(a)** $(x + 4)^2 + (y + 1)^2 = 16$; **(b)** $(x + 4)^2 + (y + 1)^2 = 7$

Do yourself - 4

1. $y = 0$ **2.** $8\sqrt{5}x + 18y = 75\sqrt{5}$ **3.** $\frac{a^2}{\ell^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$

4. $y = x \pm 7$

Do yourself - 5

1. $5x + 3y = 16$ **2.** 20 & 24 **3.** $(x^2 + y^2)^2 = 16x^2 - 9y^2$

4. $(2x + 3y - 8)(3x + 2y - 7) = 154$ **5.** $\frac{(x-6)^2}{a^2} - \frac{(y-1)^2}{b^2} = 1$ and $(x-6)^2 + (y-1)^2 = 4$

6. $3\sqrt{3}x - 13y + 15\sqrt{3} = 0$

Do yourself - 6

1. $\Delta \neq 0, h^2 > ab, a + b = 0$ **2.** $2x + y = 4$ **3.** $e = \sqrt{3}$

4. Eccentricity $= \sqrt{2}$,

Asymptotes: $x = 0$ & $y = 0$,

Transverse axis $y = -x$, Conjugate axis: $y = x$,

Centre (0,0),

Vertices (6, –6) & (–6, 6),

Foci $\left(6\sqrt{2}, -6\sqrt{2}\right)$ & $\left(-6\sqrt{2}, 6\sqrt{2}\right)$,

Length of LR $= 12\sqrt{2}$,

Equation of auxiliary circle $x^2 + y^2 = 72$,

Equation of director circle $x^2 + y^2 = 0$,

Equation of directrices $x - y = \pm\ 6\sqrt{2}$.

5. (i) $4x + y - 16 = 0$ (ii) $3x + 2y - 32 = 0$

(iii) $6x + 5y = 60$ (iv) $2x - y = 3\sqrt{8}$ and $2x - y = -3\sqrt{8}$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	B	B	A	C	B	A	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	B	D	B	A	A	C	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	C	D	B	C	C	B	D	A

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	C,D	B,C	C,D	A,B,C,D	A,C,D	B,C,D	B,C	A,B,C	B,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,D	A,B	A,B	A,B,C,D	A,B,C,D	A,C	A,B,C	A,B,C	B,C	B,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	B	A	D	A,B,C,D	A,B,C,D	A,D	0.25	2.00	C
Q.10	A	B	C	D					
	Q	S	S	R					

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	2	8	4	9	53.00	41.00	81.00	25.00	45.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	4	3	4	4	1	4	3	1
Que.	11	12	13	14	15	16	17	18		
Ans.	4	1	3	4	2	2	C	D		

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7
Ans.	B,C,D	D	A	D	B	A,D	7